

Pressure Booster System

**Hyamat V**

**Type Series Booklet**



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Type Series Booklet Hyamat V

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## Building Services: Water Supply

### Pressure Booster Systems

## Hyamat V



### Main applications

- Pressure boosting

### Fluids handled

Pump for handling clean fluids not chemically and mechanically aggressive to the pump materials.

- Drinking water
- Service water
- Cooling water

### Operating data

Operating properties

| Characteristic     |                 | Value                                                                       |
|--------------------|-----------------|-----------------------------------------------------------------------------|
| Flow rate          | Q               | Up to 660 m <sup>3</sup> /h (183 l/s with a max. of 6 pumps <sup>1)</sup> ) |
| Head               | H               | Up to 160 m                                                                 |
| Fluid temperature  | t               | Up to +70 °C,<br>up to 25 °C to DIN 1988 (DVGW)                             |
| Operating pressure | p <sub>d</sub>  | Up to 16 bar                                                                |
| Inlet pressure     | p <sub>in</sub> | Up to 10 bar                                                                |

### Designation

Example: Hyamat V 4 / 1508B / 1,2 - 3,2

Key to the designation

| Code   | Description                         |
|--------|-------------------------------------|
| Hyamat | Pressure booster system             |
| V      | One variable-speed pump             |
| 4      | Number of pumps                     |
| 15     | Pump size                           |
| 08     | Number of pump stages               |
| B      | Design status                       |
| 1,2    | Minimum inlet pressure [bar]        |
| 3,2    | Maximum usable inlet pressure [bar] |

### Design details

#### Design

- Fully automatic pressure booster package system
- Long-coupled design
- Two to six vertical high-pressure centrifugal pumps, one of them with continuously variable speed adjustment
- Hydraulic components made of stainless steel / brass
- One check valve and shut-off valves to DIN/DVGW for each pump
- Anti-vibration mounts for each pump for systems with Movitec 2B, 4B, 6B, 10B, 15B
- Systems with level-adjustable feet with rubber pads (supplied but not fitted) for systems with Movitec 25B, 45, 65 and 90B
- Membrane-type accumulator (direct-flow) to DIN 4807-5 on the discharge side, approved for drinking water
- Pressure gauge for pressure indication
- Pressure transmitter on the discharge side
- Design and function as per DIN EN 806-2, DIN 1988-500

### Installation type

- Stationary installation

### Drive

- Electric motor 50 Hz, 2-pole, IE2, special KSB model, for three-phase mains

### Automation

- Control cabinet IP 54
- Fully electronic control to ensure the required supply pressure
- LEDs indicating operational availability and fault of the system
- Service interface for connection to a PC
- Transformer for control voltage
- Motor protection switch per pump
- Lockable master switch (repair switch)
- Manual emergency switch, installed in cabinet
- Pressure transmitter on the discharge side
- Wiring plan to VDE and parts list for electric parts

<sup>1)</sup> With stand-by pump as peak load pump

- Terminal strip/terminals with identification for all connections
- Terminal connection for digital dry running protection

### Configuration and function



Hyamat V

|   |              |   |                 |
|---|--------------|---|-----------------|
| 1 | Control unit | 2 | Control cabinet |
| 3 | Pump         | 4 | Manifold        |
| 5 | Baseplate    |   |                 |

### Design

The fully automatic pressure booster system is equipped with two to six vertical high-pressure pumps (3) (one of which is speed-controlled) for pumping the fluid handled to the consumer installations in the set pressure range.

### Function

#### Automatic mode

Two to six pumps (3) are controlled and monitored by a micro-processor control unit (1). One pump is connected to a frequency inverter and controlled by the control unit so as to ensure a constant discharge pressure of the pressure booster system.

As the demand increases or decreases, peak load pumps are started and stopped automatically.

As soon as the demand increases again after one pump has been stopped, another pump which has not been in operation before is started up.

When the last (FI-controlled) pump has been stopped and the demand increases again, the next pump in line is started up and operated on the frequency inverter.

The stand-by pump is also included in the alternating cycle. The standard setting is for the pressure booster system to start automatically as a function of pressure; the actual pressure is measured by an analog pressure measuring device (pressure transmitter). The function of this pressure transmitter is monitored (live-zero).

As long as the pressure booster system is in operation, the pumps are started and stopped as a function of demand (standard setting). In this way it is ensured that the individual pumps operate only in line with the actual demand.

The variable-speed pump reduces wear as well as the pumps' frequency of starts in parallel operation. If a duty pump fails, the next pump is started up immediately and a fault is output, which can be reported via volt-free contacts (e.g. to the control station).

If the demand drops towards 0, the pressure booster system slowly runs down to the stop point. The operating status is displayed via LEDs.

### Function

#### Response to FI fault

The pressure booster system's response to a potential frequency inverter failure can be set at the control unit. As a standard, the control unit will switch to cascade operation if the frequency inverter fails. All the parameters required for this mode of operation can be set at the control unit. If automatic switchover to cascade operation is not wanted, parameters can be set for the pressure booster system to stop.

### Function

#### Energy-saving mode

In conjunction with a very large discharge-side accumulator, the energy-saving mode prevents the pressure booster system from running at the least efficient operating point, supplying very small amounts of water.

If very small amounts of water are consumed the pressure booster system only fills the downstream accumulator and stops.

Any small water volumes required can then be supplied from the accumulator.

### Function of dynamic pressure compensation

A dynamic flow rate-dependent set value re-adjustment, also called dynamic pressure compensation function, can be activated, if required.

In this mode, the system responds to decreasing flow rates by automatically increasing the set value to compensate the pressure losses in the piping.

### Function

#### Manual mode

Depending on the equipment the pressure booster system is supplied with, the pumps can be operated in manual mode in either one or two different ways.

**Standard:** By making the appropriate settings at the display, one of the pumps can be operated directly via the mains for 10 seconds, independently of the control unit. The pump will then automatically return to OFF mode.

**Supplementary equipment:** Manual-0-automatic selector switches can be supplied as supplementary equipment. They can be used to operate each pump directly on mains power, independently of the control unit.

In manual mode, a minimum flow (see table below) is essential to prevent the fluid handled and the pump from overheating when no water is consumed at the consumer installations.

### Minimum flow for pump in manual mode

Minimum flow per pump in manual mode

| Pump        | Minimum flow per pump in manual mode [l/h] |
|-------------|--------------------------------------------|
| Movitec 2B  | 200                                        |
| Movitec 4B  | 400                                        |
| Movitec 6B  | 600                                        |
| Movitec 10B | 1100                                       |
| Movitec 15B | 1600                                       |
| Movitec 25B | 2800                                       |
| Movitec 45  | 4600                                       |
| Movitec 65  | 6100                                       |
| Movitec 90B | 8500                                       |

### Example

An open 1/2-inch tap equals a water consumption of approx. 800 to 1,200 l/h.

### Dry running protection (supplementary equipment)

To protect the system from dry running, a range of protective equipment (see Supplementary equipment / Accessories) is available for various installation conditions.

Digital or analog lack-of-water monitoring equipment can be connected to the corresponding terminals.

### Field bus connection (supplementary equipment)

For remote monitoring of all system-relevant parameters and connection to a control station the system can optionally be supplied fitted with a field bus module.

### Materials

Overview of available materials

| Component               | Material                                                                                              |
|-------------------------|-------------------------------------------------------------------------------------------------------|
| Pump casing             | Stainless steel                                                                                       |
| Shroud                  | Stainless steel                                                                                       |
| Hydraulic system        | Stainless steel                                                                                       |
| Mechanical seal         | Complies with EN 12756                                                                                |
| Primary ring            | Silicon carbide                                                                                       |
| Mating ring             | Hard carbon                                                                                           |
| Elastomer               | EPDM                                                                                                  |
| Baseplate               | Steel, with powder or paint coating                                                                   |
| <b>Hydraulic design</b> |                                                                                                       |
| Manifold                | Stainless steel                                                                                       |
| Valves                  | Copper-base alloy / brass or nodular cast iron / EPDM<br>DVGW-approved<br>Approved for drinking water |
| Accumulator             | Connection made of stainless steel, flow through valve to DIN 4807-5                                  |
| Membrane                | Approved for drinking water                                                                           |

### Product benefits

- Ready-to-connect, supplied pre-set and tested for functionality
- User-friendly, straightforward menu navigation
- Reliable operation by corrosion-resistant internal parts
- Suitable for drinking water installations, manufactured under stringent hygienic conditions
- Energy-efficiency optimised by energy-saving function

## Selection information

### Requirements:

Flow rate 10 m<sup>3</sup>/h

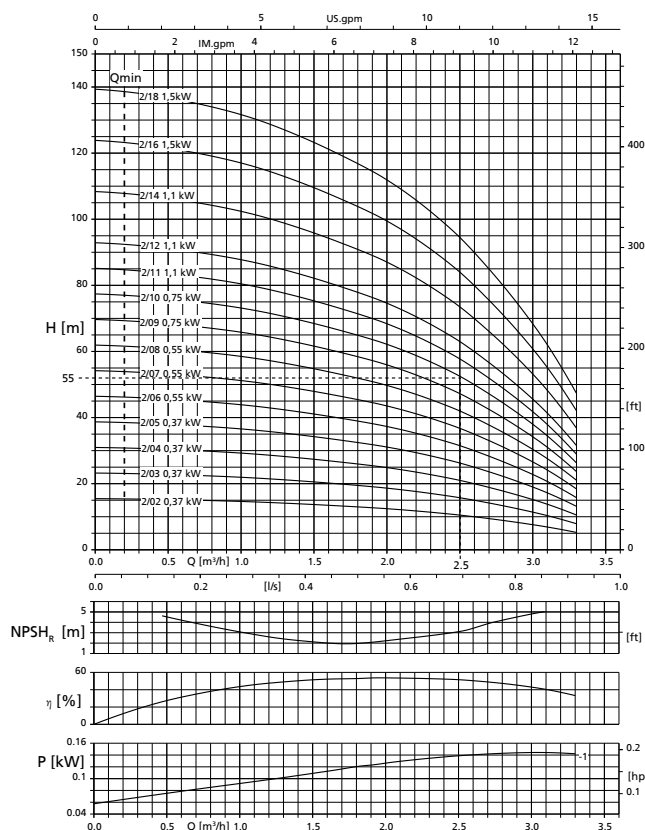
Start-up pressure 5.5 bar

Stand-by pump to DIN 1988

### Solution:

Hyamat V 5/0211 B

1. According to the table *Flow rate as a function of the number of pumps* the system may comprise 4 or 5 duty pumps (a stand-by pump has been requested).
2. According to the table *Flow rate as a function of the number of pumps* the flow rate requirement can be either 10 m<sup>3</sup>/h (4 duty pumps) or 12.5 m<sup>3</sup>/h (5 duty pumps).
3. The characteristic curves suggest Hyamat V 5/0211 B (operating point close to Q<sub>opt</sub>)



The flow rate in the characteristic curve is based on one duty pump:

The flow rate of a stand-by pump, if any, is not taken into account when calculating the flow rate required.

### Flow rate as a function of the number of pumps

| Number of duty pumps | With stand-by pump | Flow rate indicated in the diagram                                      |
|----------------------|--------------------|-------------------------------------------------------------------------|
| 1                    | No                 | Flow rate [Q] as shown in characteristic curve                          |
| 1                    | Yes                | Flow rate [Q] as shown in characteristic curve                          |
| 2                    | No                 | Flow rate required divided by 2 = flow rate in characteristic curve [Q] |
| 2                    | Yes                | Flow rate required divided by 2 = flow rate in characteristic curve [Q] |
| 3                    | No                 | Flow rate required divided by 3 = flow rate in characteristic curve [Q] |
| 3                    | Yes                | Flow rate required divided by 3 = flow rate in characteristic curve [Q] |
| 4                    | No                 | Flow rate required divided by 4 = flow rate in characteristic curve [Q] |
| 4                    | Yes                | Flow rate required divided by 4 = flow rate in characteristic curve [Q] |
| 5                    | No                 | Flow rate required divided by 5 = flow rate in characteristic curve [Q] |
| 5                    | Yes                | Flow rate required divided by 5 = flow rate in characteristic curve [Q] |
| 6                    | No                 | Flow rate required divided by 6 = flow rate in characteristic curve [Q] |

**Technical data**
**Electrical performance data**

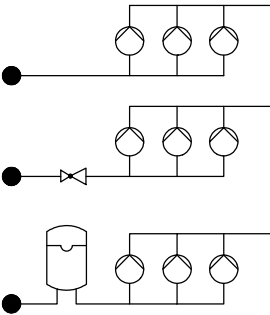
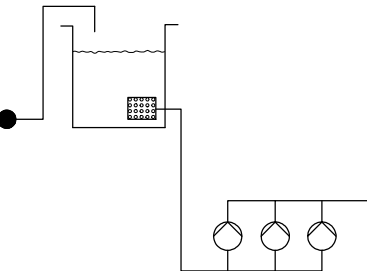
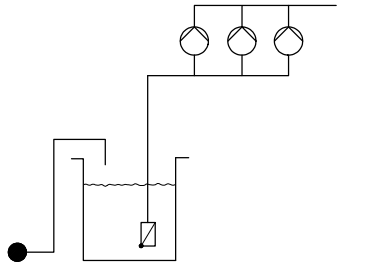
Electrical performance data

| Hyamat V<br>with Movitec<br>pumps | Rated power per motor<br>[kW] | Rated current per motor at 400<br>V<br>[A] | Total rated power [kVA]  |      |      |      |      |
|-----------------------------------|-------------------------------|--------------------------------------------|--------------------------|------|------|------|------|
|                                   |                               |                                            | Number of pumps (motors) |      |      |      |      |
|                                   |                               |                                            | 2                        | 3    | 4    | 5    | 6    |
| 0202B                             | 0,37                          | 0,94                                       | 1,3                      | 2    | 2,6  | 3,3  | 3,9  |
| 0203B                             | 0,37                          | 0,94                                       | 1,3                      | 2    | 2,6  | 3,3  | 3,9  |
| 0204B                             | 0,37                          | 0,94                                       | 1,3                      | 2    | 2,6  | 3,3  | 3,9  |
| 0205B                             | 0,37                          | 0,94                                       | 1,3                      | 2    | 2,6  | 3,3  | 3,9  |
| 0206B                             | 0,55                          | 1,33                                       | 1,9                      | 2,8  | 3,7  | 4,7  | 5,6  |
| 0207B                             | 0,55                          | 1,33                                       | 1,9                      | 2,8  | 3,7  | 4,7  | 5,6  |
| 0208B                             | 0,55                          | 1,33                                       | 1,9                      | 2,8  | 3,7  | 4,7  | 5,6  |
| 0209B                             | 0,75                          | 1,73                                       | 2,5                      | 3,7  | 4,9  | 6,1  | 7,3  |
| 0210B                             | 0,75                          | 1,73                                       | 2,5                      | 3,7  | 4,9  | 6,1  | 7,3  |
| 0211B                             | 1,1                           | 2,39                                       | 3,4                      | 5,1  | 6,7  | 8,4  | 10   |
| 0212B                             | 1,1                           | 2,39                                       | 3,4                      | 5,1  | 6,7  | 8,4  | 10   |
| 0214B                             | 1,1                           | 2,39                                       | 3,4                      | 5,1  | 6,7  | 8,4  | 10   |
| 0216B                             | 1,5                           | 2,9                                        | 4,1                      | 6,1  | 8,1  | 10,1 | 12,2 |
| 0218B                             | 1,5                           | 2,9                                        | 4,1                      | 6,1  | 8,1  | 10,1 | 12,2 |
|                                   |                               |                                            |                          |      |      |      |      |
| 0402B                             | 0,37                          | 0,94                                       | 1,3                      | 2    | 2,6  | 3,3  | 3,9  |
| 0403B                             | 0,55                          | 1,33                                       | 1,9                      | 2,8  | 3,7  | 4,7  | 5,6  |
| 0404B                             | 0,55                          | 1,33                                       | 1,9                      | 2,8  | 3,7  | 4,7  | 5,6  |
| 0405B                             | 0,75                          | 1,73                                       | 2,5                      | 3,7  | 4,9  | 6,1  | 7,3  |
| 0406B                             | 1,1                           | 2,39                                       | 3,4                      | 5,1  | 6,7  | 8,4  | 10   |
| 0407B                             | 1,1                           | 2,39                                       | 3,4                      | 5,1  | 6,7  | 8,4  | 10   |
| 0408B                             | 1,5                           | 2,9                                        | 4,1                      | 6,1  | 8,1  | 10,1 | 12,2 |
| 0409B                             | 1,5                           | 2,9                                        | 4,1                      | 6,1  | 8,1  | 10,1 | 12,2 |
| 0410B                             | 1,5                           | 2,9                                        | 4,1                      | 6,1  | 8,1  | 10,1 | 12,2 |
| 0411B                             | 2,2                           | 4,17                                       | 5,9                      | 8,8  | 11,7 | 14,6 | 17,5 |
| 0412B                             | 2,2                           | 4,17                                       | 5,9                      | 8,8  | 11,7 | 14,6 | 17,5 |
| 0414B                             | 2,2                           | 4,17                                       | 5,9                      | 8,8  | 11,7 | 14,6 | 17,5 |
| 0416B                             | 3                             | 5,55                                       | 7,9                      | 11,7 | 15,6 | 19,4 | 23,3 |
|                                   |                               |                                            |                          |      |      |      |      |
| 0602B                             | 0,37                          | 0,94                                       | 1,3                      | 2    | 2,6  | 3,3  | 3,9  |
| 0603B                             | 0,75                          | 1,73                                       | 2,5                      | 3,7  | 4,9  | 6,1  | 7,3  |
| 0604B                             | 1,1                           | 2,39                                       | 3,4                      | 5,1  | 6,7  | 8,4  | 10   |
| 0605B                             | 1,1                           | 2,39                                       | 3,4                      | 5,1  | 6,7  | 8,4  | 10   |
| 0606B                             | 1,5                           | 2,9                                        | 4,1                      | 6,1  | 8,1  | 10,1 | 12,2 |
| 0607B                             | 1,5                           | 2,9                                        | 4,1                      | 6,1  | 8,1  | 10,1 | 12,2 |
| 0608B                             | 2,2                           | 4,17                                       | 5,9                      | 8,8  | 11,7 | 14,6 | 17,5 |
| 0609B                             | 2,2                           | 4,17                                       | 5,9                      | 8,8  | 11,7 | 14,6 | 17,5 |
| 0610B                             | 2,2                           | 4,17                                       | 5,9                      | 8,8  | 11,7 | 14,6 | 17,5 |
| 0611B                             | 3                             | 5,55                                       | 7,9                      | 11,7 | 15,6 | 19,4 | 23,3 |
| 0612B                             | 3                             | 5,55                                       | 7,9                      | 11,7 | 15,6 | 19,4 | 23,3 |
| 0614B                             | 3                             | 5,55                                       | 7,9                      | 11,7 | 15,6 | 19,4 | 23,3 |
|                                   |                               |                                            |                          |      |      |      |      |
| 1002B                             | 0,75                          | 1,73                                       | 2,5                      | 3,7  | 4,9  | 6,1  | 7,3  |
| 1003B                             | 1,1                           | 2,39                                       | 3,4                      | 5,1  | 6,7  | 8,4  | 10   |
| 1004B                             | 1,5                           | 2,9                                        | 4,1                      | 6,1  | 8,1  | 10,1 | 12,2 |
| 1005B                             | 2,2                           | 4,17                                       | 5,9                      | 8,8  | 11,7 | 14,6 | 17,5 |
| 1006B                             | 2,2                           | 4,17                                       | 5,9                      | 8,8  | 11,7 | 14,6 | 17,5 |
| 1007B                             | 3                             | 5,55                                       | 7,9                      | 11,7 | 15,6 | 19,4 | 23,3 |
| 1008B                             | 3                             | 5,55                                       | 7,9                      | 11,7 | 15,6 | 19,4 | 23,3 |
| 1009B                             | 4                             | 7,29                                       | 10,4                     | 15,4 | 20,5 | 25,5 | 30,6 |
| 1010B                             | 4                             | 7,29                                       | 10,4                     | 15,4 | 20,5 | 25,5 | 30,6 |
| 1011B                             | 4                             | 7,29                                       | 10,4                     | 15,4 | 20,5 | 25,5 | 30,6 |
| 1013B                             | 5,5                           | 10,07                                      | 14,3                     | 21,3 | 28,3 | 35,2 | 42,2 |
|                                   |                               |                                            |                          |      |      |      |      |
| 1502B                             | 2,2                           | 4,17                                       | 5,9                      | 8,8  | 11,7 | 14,6 | 17,5 |
| 1503B                             | 3                             | 5,55                                       | 7,9                      | 11,7 | 15,6 | 19,4 | 23,3 |

| Hyamat V<br>with Movitec<br>pumps | Rated power per motor | Rated current per motor at 400<br>V | Total rated power [kVA]  |      |       |       |       |
|-----------------------------------|-----------------------|-------------------------------------|--------------------------|------|-------|-------|-------|
|                                   |                       |                                     | Number of pumps (motors) |      |       |       |       |
|                                   | [kW]                  | [A]                                 | 2                        | 3    | 4     | 5     | 6     |
| 1504B                             | 4                     | 7,29                                | 10,4                     | 15,4 | 20,5  | 25,5  | 30,6  |
| 1505B                             | 5,5                   | 10,07                               | 14,3                     | 21,3 | 28,3  | 35,2  | 42,2  |
| 1506B                             | 5,5                   | 10,07                               | 14,3                     | 21,3 | 28,3  | 35,2  | 42,2  |
| 1507B                             | 7,5                   | 13,6                                | 19,3                     | 28,7 | 38,2  | 47,6  | 57    |
| 1508B                             | 7,5                   | 13,6                                | 19,3                     | 28,7 | 38,2  | 47,6  | 57    |
|                                   |                       |                                     |                          |      |       |       |       |
| 2502B                             | 4                     | 7,29                                | 10,4                     | 15,4 | 20,5  | 25,5  | 30,6  |
| 2503B                             | 5,5                   | 10,07                               | 14,3                     | 21,3 | 28,3  | 35,2  | 42,2  |
| 2504B                             | 7,5                   | 13,6                                | 19,3                     | 28,7 | 38,2  | 47,6  | 57    |
| 2505B                             | 11                    | 19,47                               | 27,7                     | 41,1 | 54,6  | 68,1  | 81,6  |
| 2506B                             | 11                    | 19,47                               | 27,7                     | 41,1 | 54,6  | 68,1  | 81,6  |
| 2507B                             | 15                    | 26,7                                | 37,9                     | 56,4 | 74,9  | 93,4  | 111,9 |
|                                   |                       |                                     |                          |      |       |       |       |
| 4502-1                            | 5,5                   | 14,4                                | 20,5                     | 30,4 | 40,4  | 50,4  | 60,4  |
| 4502                              | 7,5                   | 17,6                                | 25                       | 37,2 | 49,4  | 61,6  | 73,8  |
| 4503-1                            | 11                    | 27,3                                | 38,8                     | 57,7 | 76,6  | 95,5  | 114,4 |
| 4503                              | 11                    | 27,3                                | 38,8                     | 57,7 | 76,6  | 95,5  | 114,4 |
| 4504-1                            | 11                    | 27,3                                | 38,8                     | 57,7 | 76,6  | 95,5  | 114,4 |
| 4504                              | 15                    | 32                                  | 45,4                     | 67,6 | 89,8  | 112   | 134,1 |
| 4505-1                            | 15                    | 32                                  | 45,4                     | 67,6 | 89,8  | 112   | 134,1 |
| 4505                              | 18,5                  | 42,2                                | 59,9                     | 89,2 | 118,4 | 147,6 | 176,9 |
| 4506-1                            | 18,5                  | 42,2                                | 59,9                     | 89,2 | 118,4 | 147,6 | 176,9 |
| 4506                              | 22                    | 46,3                                | 65,8                     | 97,8 | 129,9 | 162   | 194,1 |
|                                   |                       |                                     |                          |      |       |       |       |
| 6502                              | 5,5                   | 14,4                                | 20,5                     | 30,4 | 40,4  | 50,4  | 60,4  |
| 6503                              | 11                    | 27,3                                | 38,8                     | 57,7 | 76,6  | 95,5  | 114,4 |
| 6504                              | 11                    | 27,3                                | 38,8                     | 57,7 | 76,6  | 95,5  | 114,4 |
| 6505                              | 15                    | 32                                  | 45,4                     | 67,6 | 89,8  | 112   | 134,1 |
| 6506                              | 15                    | 32                                  | 45,4                     | 67,6 | 89,8  | 112   | 134,1 |
| 6507                              | 18,5                  | 42,2                                | 59,9                     | 89,2 | 118,4 | 147,6 | 176,9 |
| 6508                              | 22                    | 46,3                                | 65,8                     | 97,8 | 129,9 | 162   | 194,1 |
|                                   |                       |                                     |                          |      |       |       |       |
| 9002-2-2B                         | 11                    | 20,3                                | 28,8                     | 42,9 | 57    | 71    | 85,1  |
| 9002-2-1B                         | 15                    | 26,7                                | 37,9                     | 56,4 | 74,9  | 93,4  | 111,9 |
| 9002-2B                           | 15                    | 26,7                                | 37,9                     | 56,4 | 74,9  | 93,4  | 111,9 |
| 9002-3-2B                         | 18,5                  | 33,7                                | 47,9                     | 71,2 | 94,6  | 117,9 | 141,3 |
| 9002-3-1B                         | 22                    | 39,8                                | 56,6                     | 84,1 | 111,7 | 139,2 | 166,8 |
| 9002-3B                           | 22                    | 39,8                                | 56,6                     | 84,1 | 111,7 | 139,2 | 166,8 |

## Type of connection

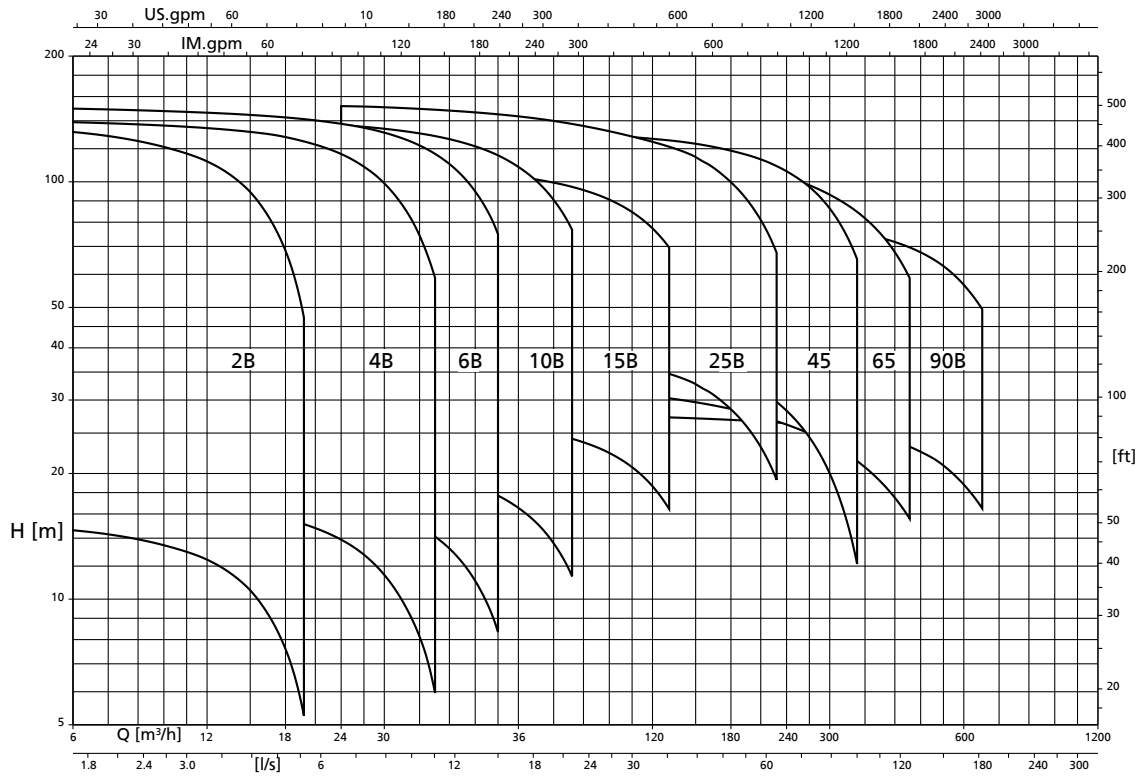
Types of connection (schematic)

| Direct                                                                                                                                                                                                                                                                                                                                                                                                       | Indirect                                                                                                                                                                                                                       |                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>1952x106</p>                                                                                                                                                                                                                                                                                                            |  <p>1952x107</p>                                                                                                                              |  <p>1952x108</p>                                                                                                 |
| <p><b>Inlet pressure monitoring (see Supplementary equipment or Accessories)</b></p> <p>At <math>p_{in} &gt; 0.5</math> bar (min. 1 bar, DIN 1988)</p> <ul style="list-style-type: none"> <li>- Pressure switch</li> <li>- Pressure sensor</li> </ul> <p>At <math>p_{in} &lt; 0.5</math> bar</p> <ul style="list-style-type: none"> <li>- Pressure sensor</li> <li>- Flow monitoring<sup>3)</sup></li> </ul> | <ul style="list-style-type: none"> <li>- Float switch</li> <li>- Set of electrodes and relay</li> <li>- Dry running protection for PE inlet tank</li> <li>- Pressure sensor</li> <li>- Flow monitoring<sup>3)</sup></li> </ul> | <ul style="list-style-type: none"> <li>- Float switch</li> <li>- Set of electrodes and relay</li> <li>- Dry running protection for PE inlet tank</li> <li>- Flow monitoring<sup>3)</sup></li> </ul> |

<sup>2)</sup> Non-priming pumps, suitable for suction lift operation (for selection, please consult KSB)

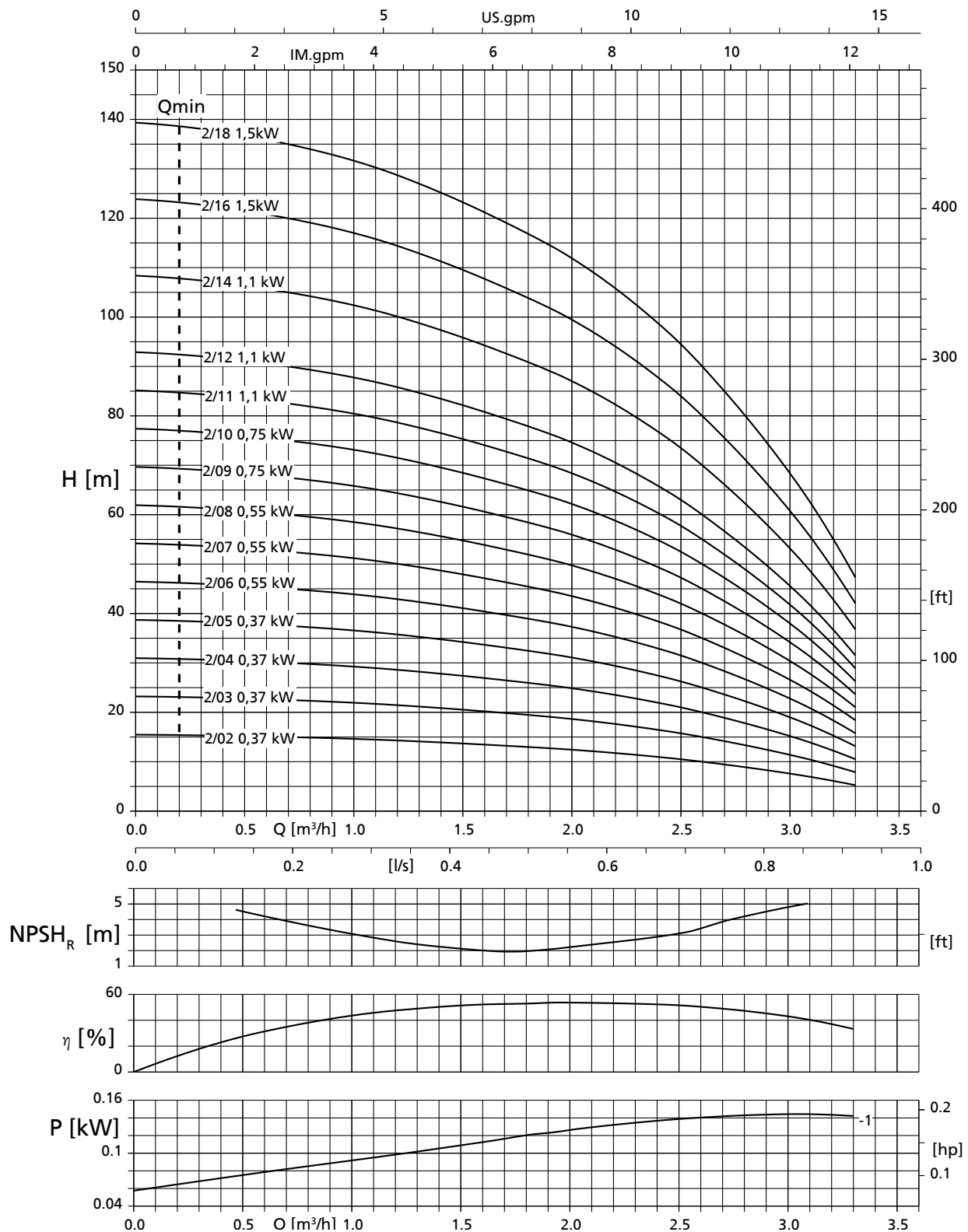
<sup>3)</sup> Note: Automatic reset is not possible for this type of dry running protection.

Hyamat K, V, VP; n = 2900 rpm



## Characteristic curves

### Hyamat K, V, VP with Movitec 2B

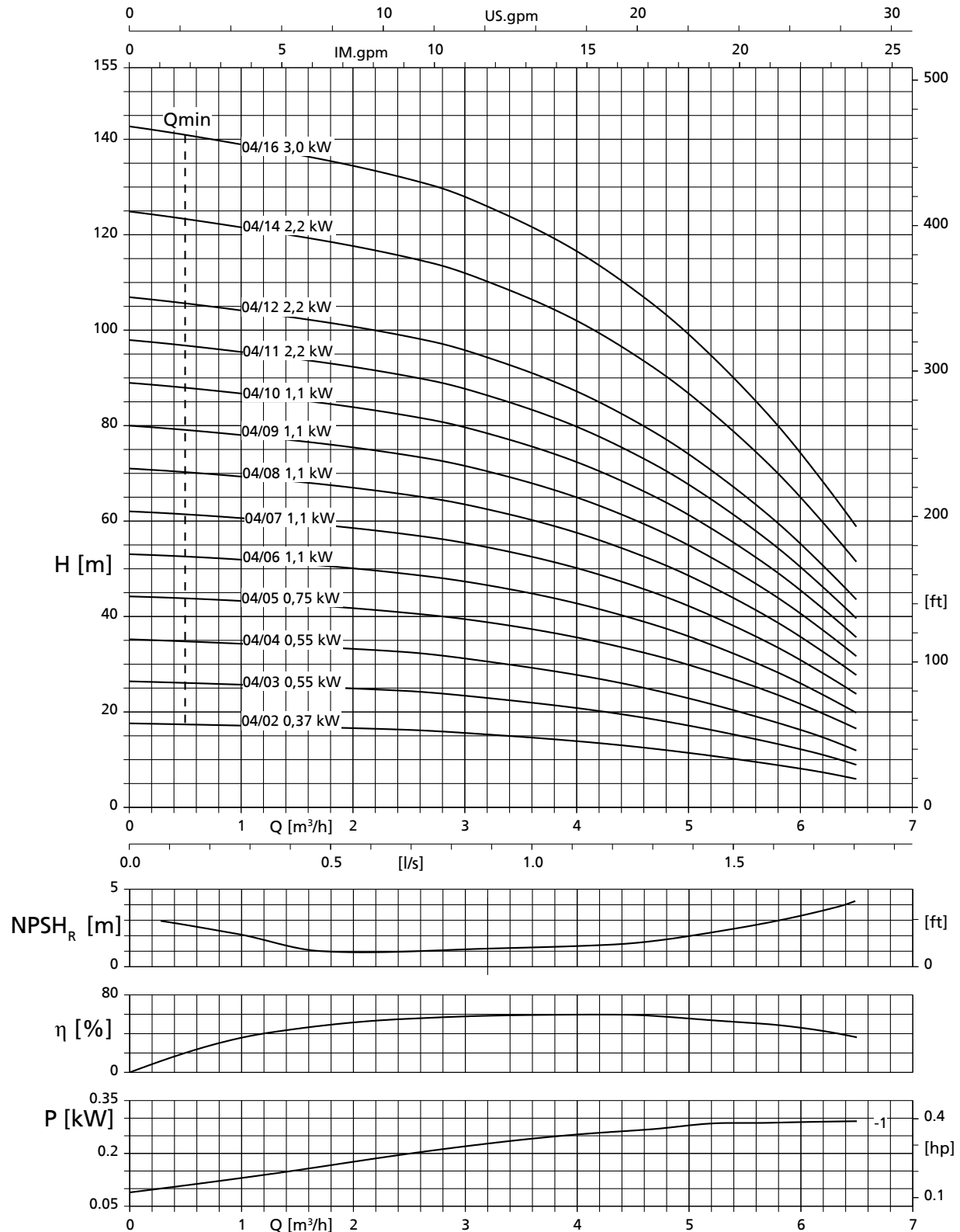


The flow rate in the characteristic curve is based on one duty pump:

The flow rate of a stand-by pump, if any, is not taken into account when calculating the flow rate required.

Flow rates for multiple pump systems (⇒ Page 7)

Hyamat K, V, VP with Movitec 4B

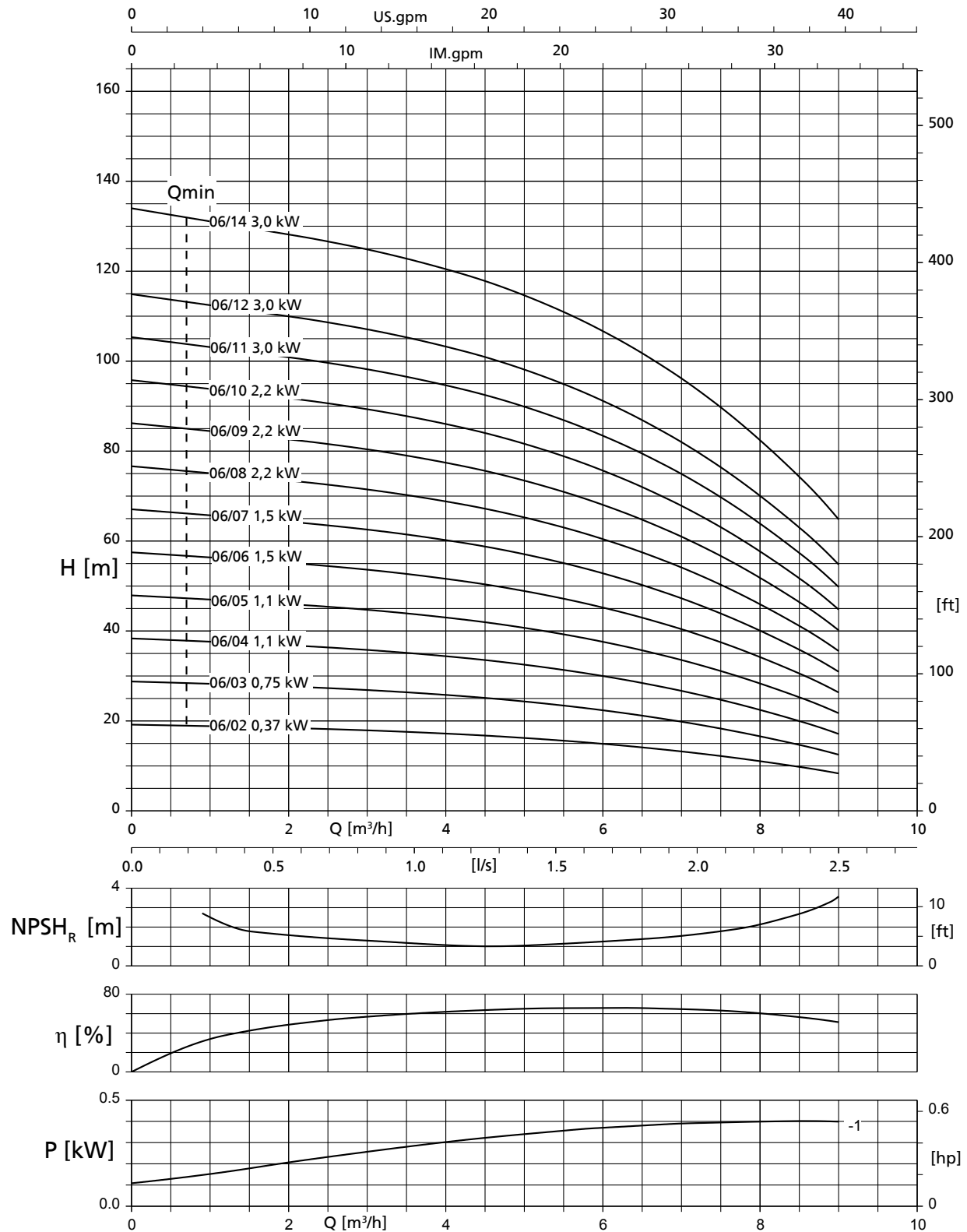


The flow rate in the characteristic curve is based on one duty pump:

The flow rate of a stand-by pump, if any, is not taken into account when calculating the flow rate required.

Flow rates for multiple pump systems (⇒ Page 7)

Hyamat K, V, VP with Movitec 6B

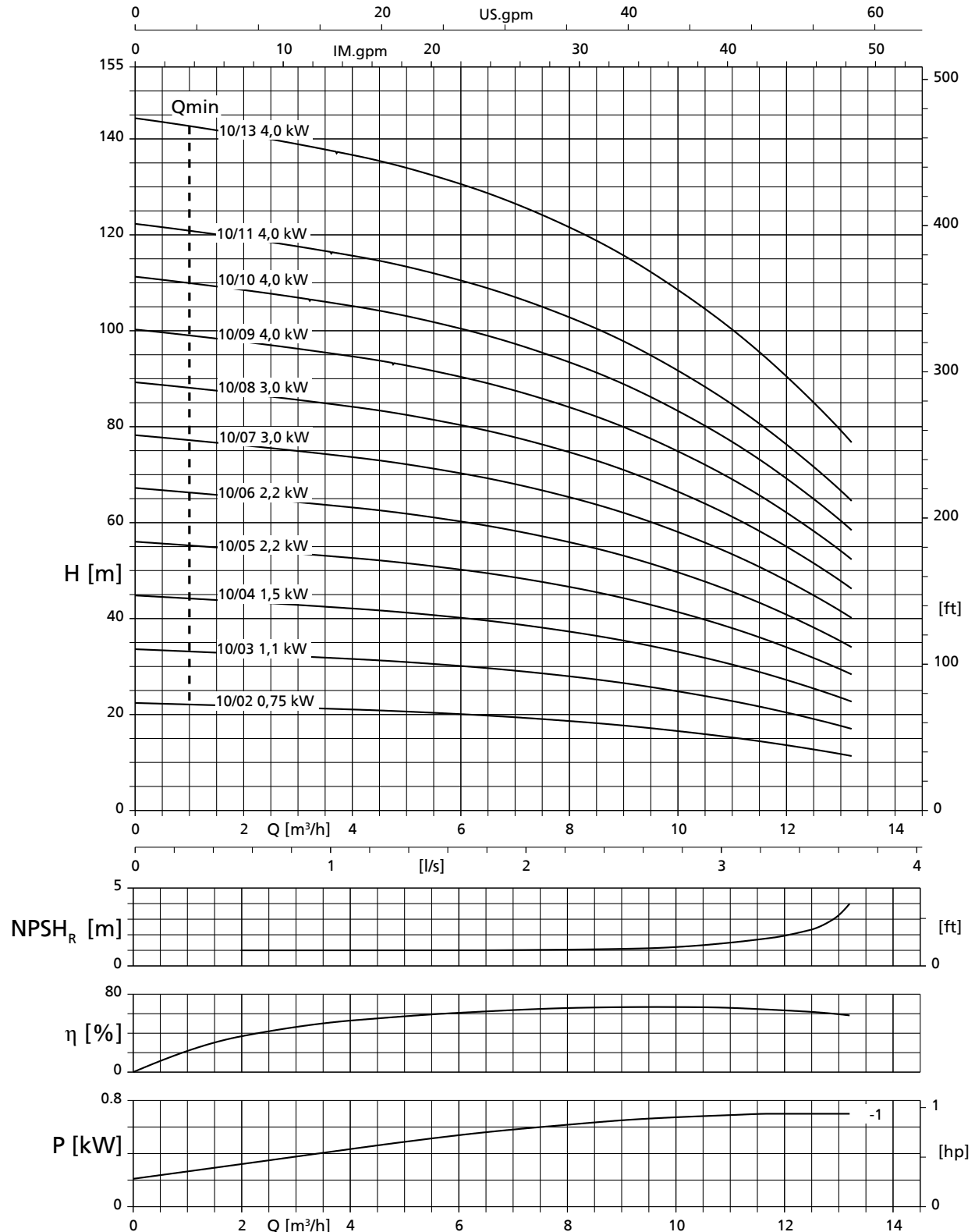


The flow rate in the characteristic curve is based on one duty pump:

The flow rate of a stand-by pump, if any, is not taken into account when calculating the flow rate required.

Flow rates for multiple pump systems (⇒ Page 7)

Hyamat K, V, VP with Movitec 10B

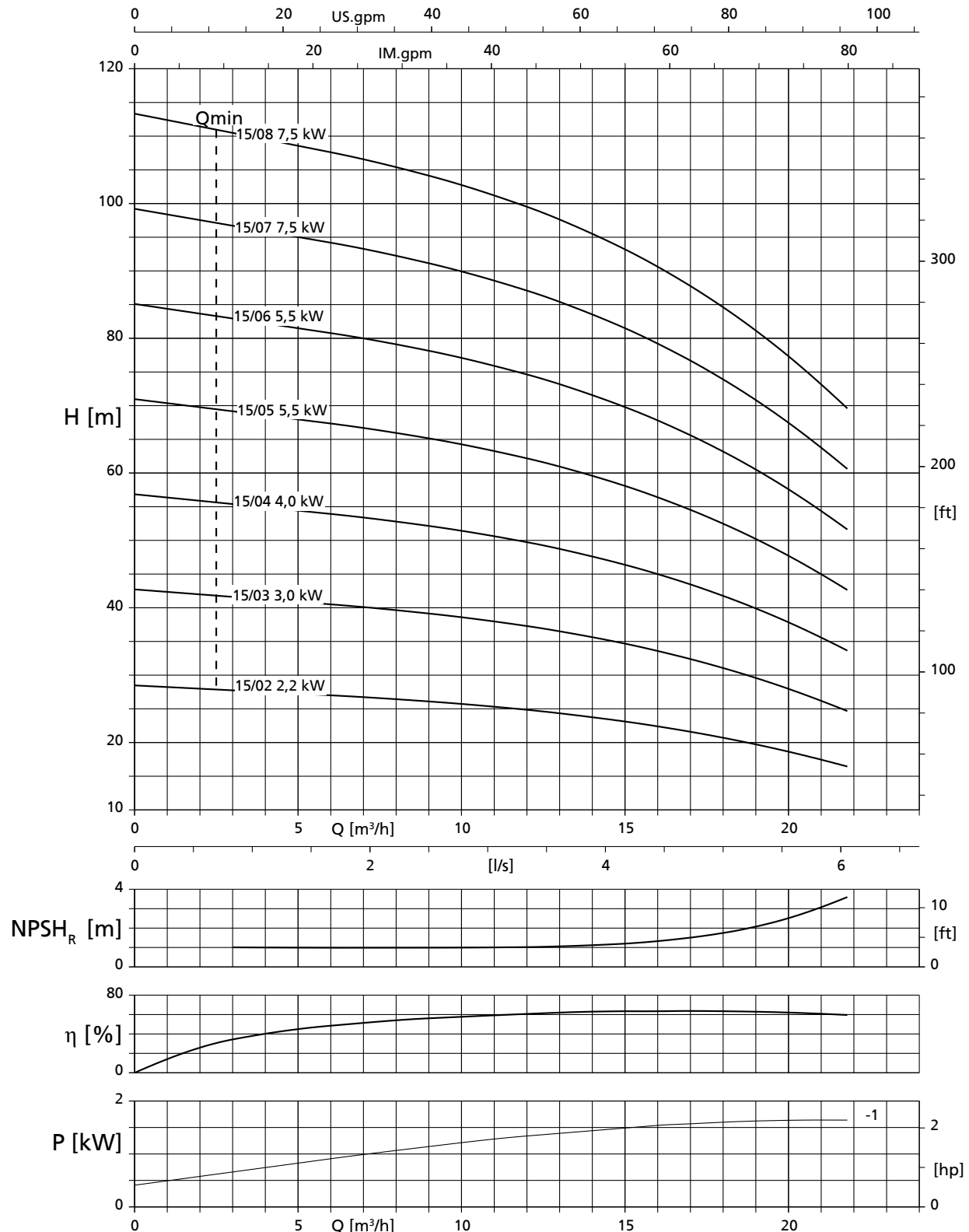


The flow rate in the characteristic curve is based on one duty pump:

The flow rate of a stand-by pump, if any, is not taken into account when calculating the flow rate required.

Flow rates for multiple pump systems (⇒ Page 7)

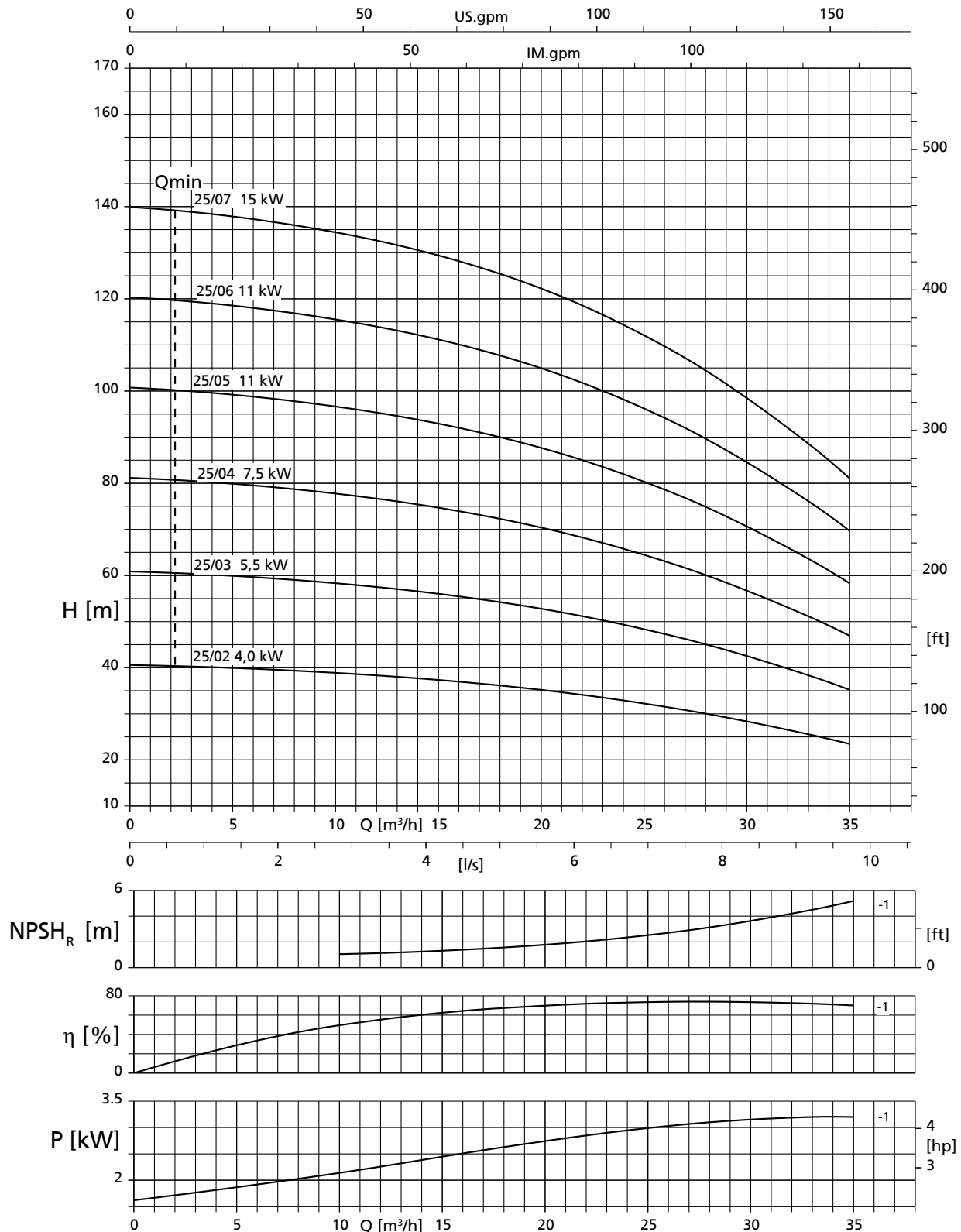
Hyamat K, V, VP with Movitec 15B



The flow rate in the characteristic curve is based on one duty pump:  
The flow rate of a stand-by pump, if any, is not taken into account when calculating the flow rate required.

Flow rates for multiple pump systems (⇒ Page 7)

Hyamat K, V, VP with Movitec 25B

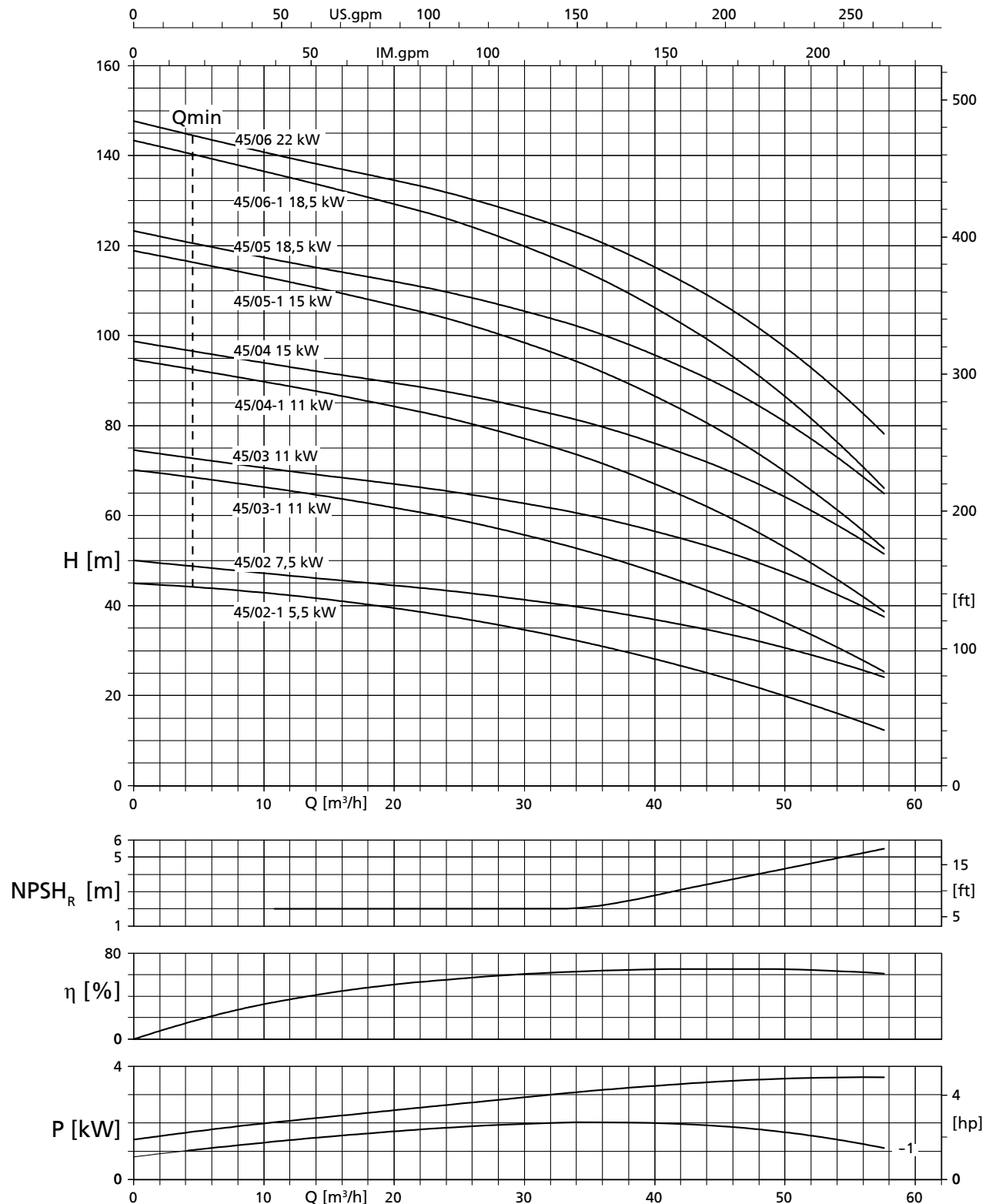


The flow rate in the characteristic curve is based on one duty pump:

The flow rate of a stand-by pump, if any, is not taken into account when calculating the flow rate required.

Flow rates for multiple pump systems (⇒ Page 7)

Hyamat K, V, VP with Movitec 45

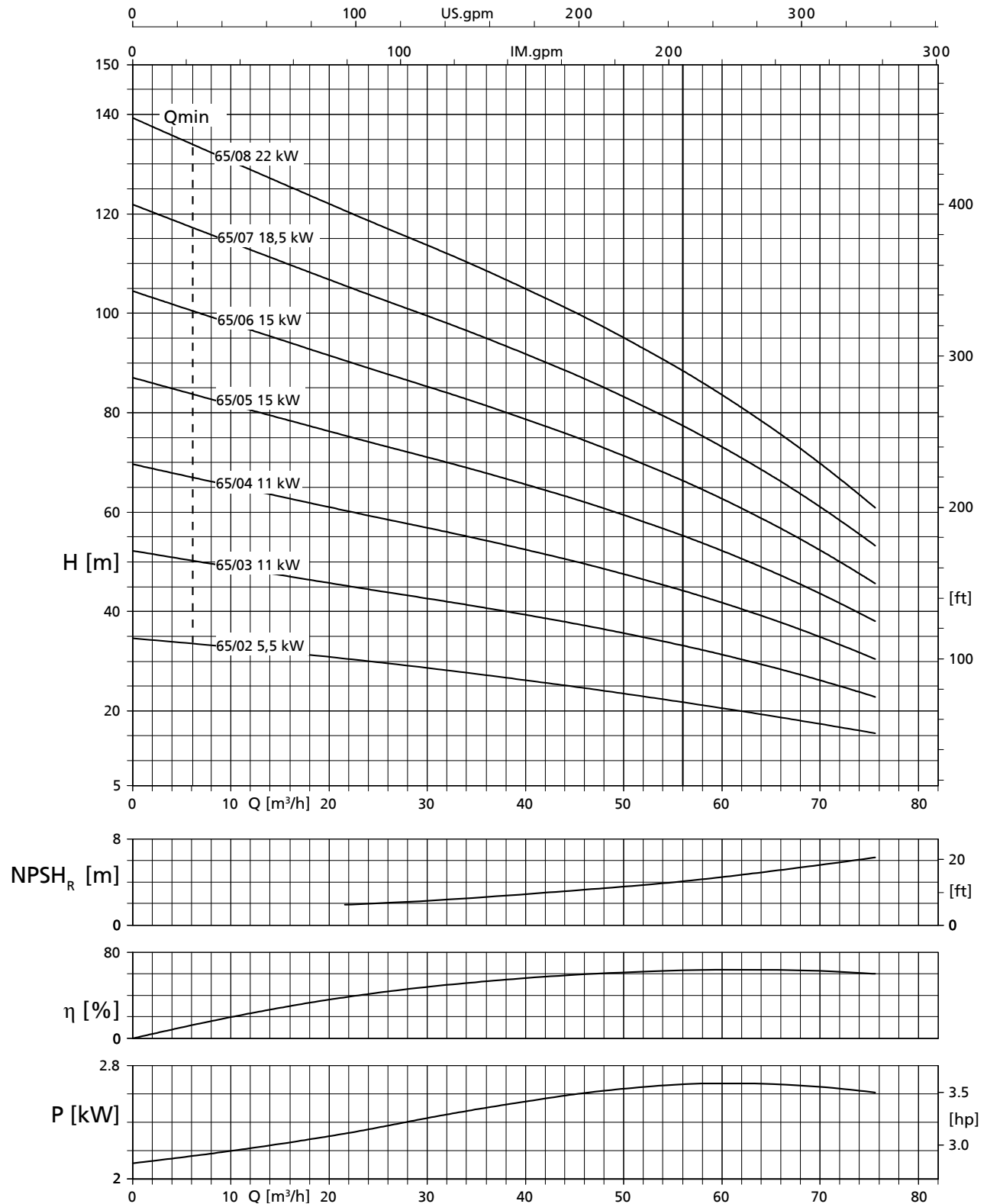


The flow rate in the characteristic curve is based on one duty pump:

The flow rate of a stand-by pump, if any, is not taken into account when calculating the flow rate required.

Flow rates for multiple pump systems (⇒ Page 7)

### Hyamat K, V, VP with Movitec 65

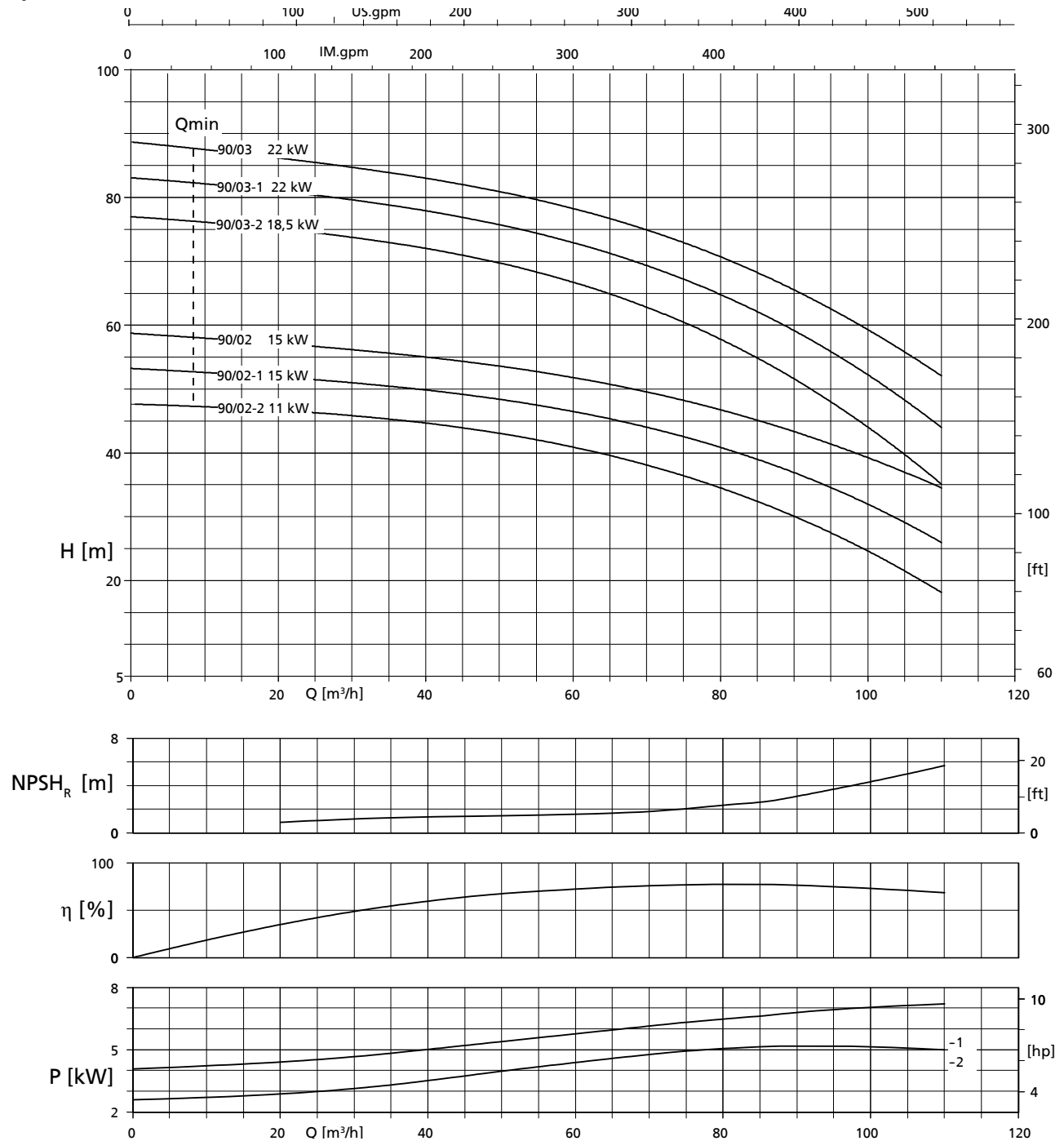


The flow rate in the characteristic curve is based on one duty pump:

The flow rate of a stand-by pump, if any, is not taken into account when calculating the flow rate required.

Flow rates for multiple pump systems (⇒ Page 7)

### Hyamat K, V, VP with Movitec 90B



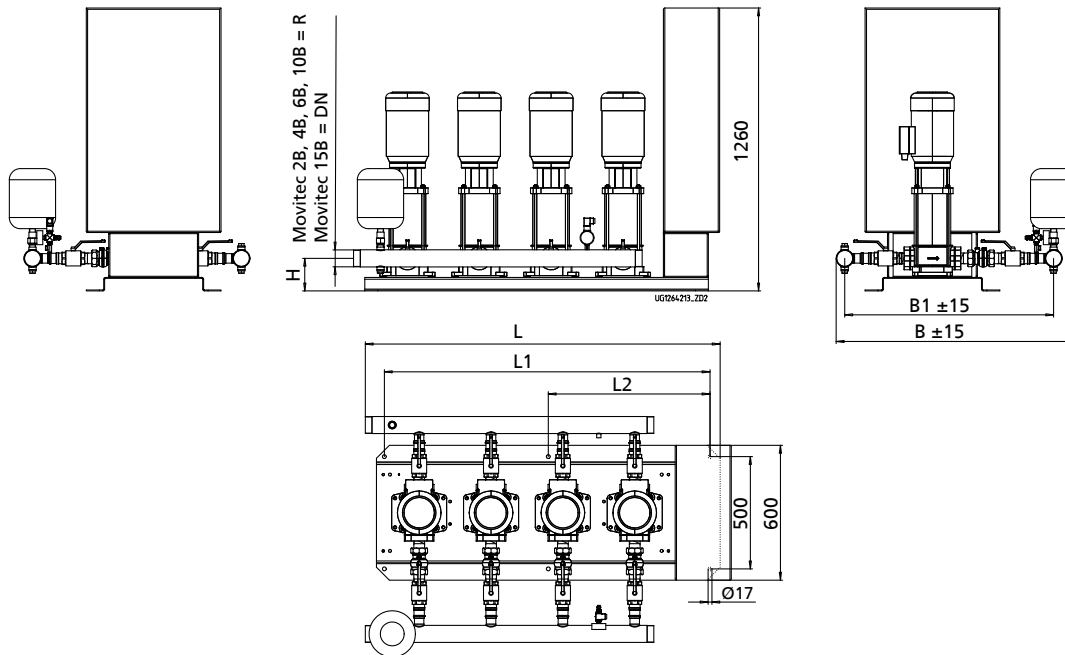
The flow rate in the characteristic curve is based on one duty pump:

The flow rate of a stand-by pump, if any, is not taken into account when calculating the flow rate required.

Flow rates for multiple pump systems (⇒ Page 7)

## Dimensions and weights

### System dimensions

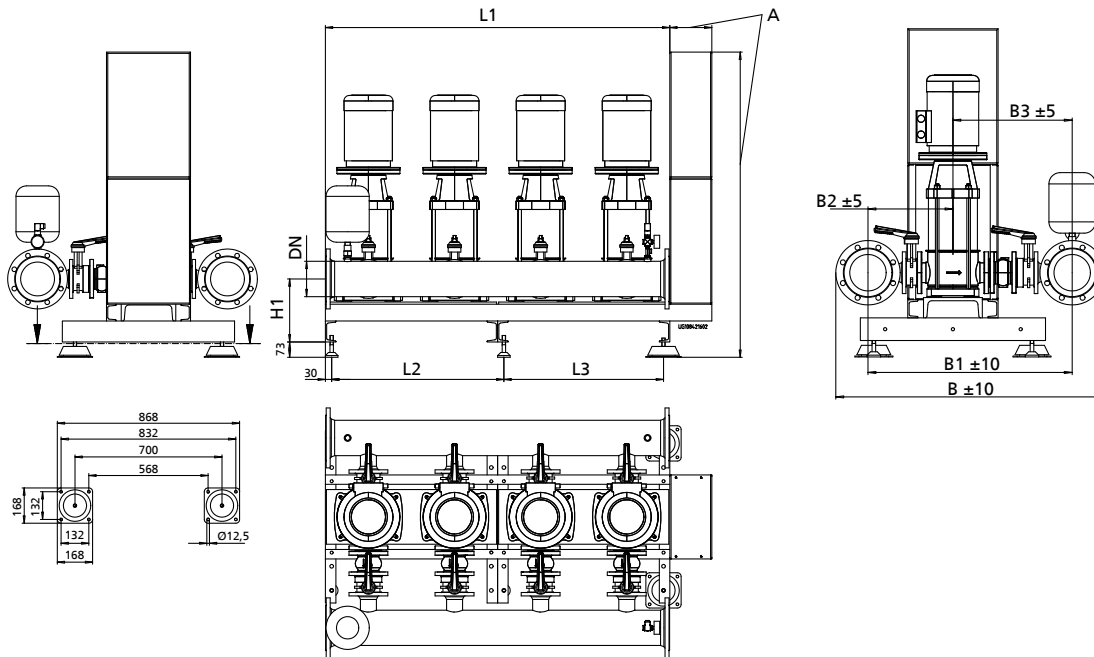


### Dimensions of Hyamat V with Movitec 2B, 4B, 6B, 10B and 15B

Dimensions of Hyamat V with Movitec 2B, 4B, 6B, 10B and 15B [mm]

| Number of pumps | 2     | 3       | 4       | 5       | 6       | Pump                    |
|-----------------|-------|---------|---------|---------|---------|-------------------------|
| B               | 896   | 896     | 896     | 920     | 920     | 2B/.. and 4B/...        |
|                 | 961   | 961     | 961     | 987     | 987     | 6B/..                   |
|                 | 1050  | 1073    | 1073    | 1073    | 1090    | 10B/..                  |
|                 | 1097  | 1097    | 1272    | 1221    | 1352    | 15B/..                  |
| B1              | 763   | 763     | 763     | 778     | 778     | 2B/.. and 4B/...        |
|                 | 828   | 828     | 828     | 846     | 846     | 6B/..                   |
|                 | 916   | 932     | 932     | 932     | 943     | 10B/..                  |
|                 | 894   | 894     | 1052    | 1001    | 1067    | 15B/..                  |
| L               | 825   | 1055    | 1285    | 1605    | 1925    | 2B/.., 4B/.. and 6B/... |
|                 | 985   | 1260    | 1580    | 1900    | 2220    | 10B/..                  |
|                 | 980   | 1210    | 1544    | 1850    | 2170    | 15B/..                  |
| L1              | 670   | 900     | 1130    | 1450    | 1770    | 2B/.., 4B/.. and 6B/... |
|                 | 900   | 1130    | 1450    | 1770    | 2090    | 10B/..                  |
|                 | 900   | 1130    | 1450    | 1770    | 2090    | 15B/..                  |
| L2              | -     | -       | 560     | 720     | 880     | 2B/.., 4B/.. and 6B/... |
|                 | -     | 560     | 720     | 880     | 1040    | 10B/..                  |
|                 | -     | 560     | 720     | 880     | 1040    | 15B/..                  |
| R               | R 2   | R 2     | R 2     | R 2 1/2 | R 2 1/2 | 2B/.., 4B/.. and 6B/... |
|                 | R 2   | R 2 1/2 | R 2 1/2 | R 2 1/2 | R 3     | 10B/..                  |
| DN              | DN 80 | DN 80   | DN 100  | DN 100  | DN 150  | 15B/..                  |
| H1              | 115   | 115     | 115     | 115     | 115     | 2B/.., 4B/.. and 6B/... |
|                 | 145   | 145     | 145     | 145     | 145     | 10B/..                  |
|                 | 145   | 145     | 145     | 155     | 155     | 15B/..                  |

**Colours:** baseplate RAL 5002, control cabinet RAL 7035



Dimensions of Hyamat V with Movitec 25B, 45, 65 and 90B

A See control cabinets (⇒ Page 23)

Dimensions of Hyamat V with Movitec 25B, 45, 65 and 90B [mm]

| Number of pumps | 2    | 3    | 4    | 5    | 6    | Pump   |
|-----------------|------|------|------|------|------|--------|
| B               | 1074 | 1074 | 1189 | 1189 | 1189 | 25B/.. |
|                 | 1094 | 1203 | 1203 | 1304 | 1304 | 45/..  |
|                 | 1320 | 1320 | 1421 | 1421 | 1421 | 65/..  |
|                 | 1335 | 1436 | 1436 | 1561 | 1561 | 90B/.. |
| B1              | 854  | 854  | 904  | 904  | 904  | 25B/.. |
|                 | 874  | 918  | 918  | 964  | 964  | 45/..  |
|                 | 1035 | 1035 | 1081 | 1081 | 1081 | 65/..  |
|                 | 1050 | 1096 | 1096 | 1156 | 1156 | 90B/.. |
| B2              | 351  | 351  | 376  | 376  | 376  | 25B/.. |
|                 | 351  | 373  | 373  | 396  | 396  | 45/..  |
|                 | 431  | 431  | 454  | 454  | 454  | 65/..  |
|                 | 439  | 462  | 462  | 492  | 492  | 90B/.. |
| B3              | 503  | 503  | 528  | 528  | 528  | 25B/.. |
|                 | 523  | 545  | 545  | 568  | 568  | 45/..  |
|                 | 604  | 604  | 627  | 627  | 627  | 65/..  |
|                 | 611  | 634  | 634  | 664  | 664  | 90B/.. |
| L1              | 820  | 1230 | 1640 | 2050 | 2460 |        |
| L2              | -    | -    | 820  | 1230 | 1230 |        |
| L3              | 760  | 1170 | 760  | 760  | 1170 |        |
| DN              | 100  | 100  | 150  | 150  | 150  | 25B/.. |
|                 | 100  | 150  | 150  | 200  | 200  | 45/..  |
|                 | 150  | 150  | 200  | 200  | 200  | 65/..  |
|                 | 150  | 200  | 200  | 250  | 250  | 90B/.. |
| H1              | 302  | 302  | 302  | 302  | 302  | 25B/.. |
|                 | 302  | 302  | 302  | 302  | 302  | 45/..  |
|                 | 337  | 337  | 337  | 337  | 337  | 65/..  |
|                 | 337  | 337  | 337  | 337  | 337  | 90B/.. |

Flanges drilled to EN 1092-1 PN 16

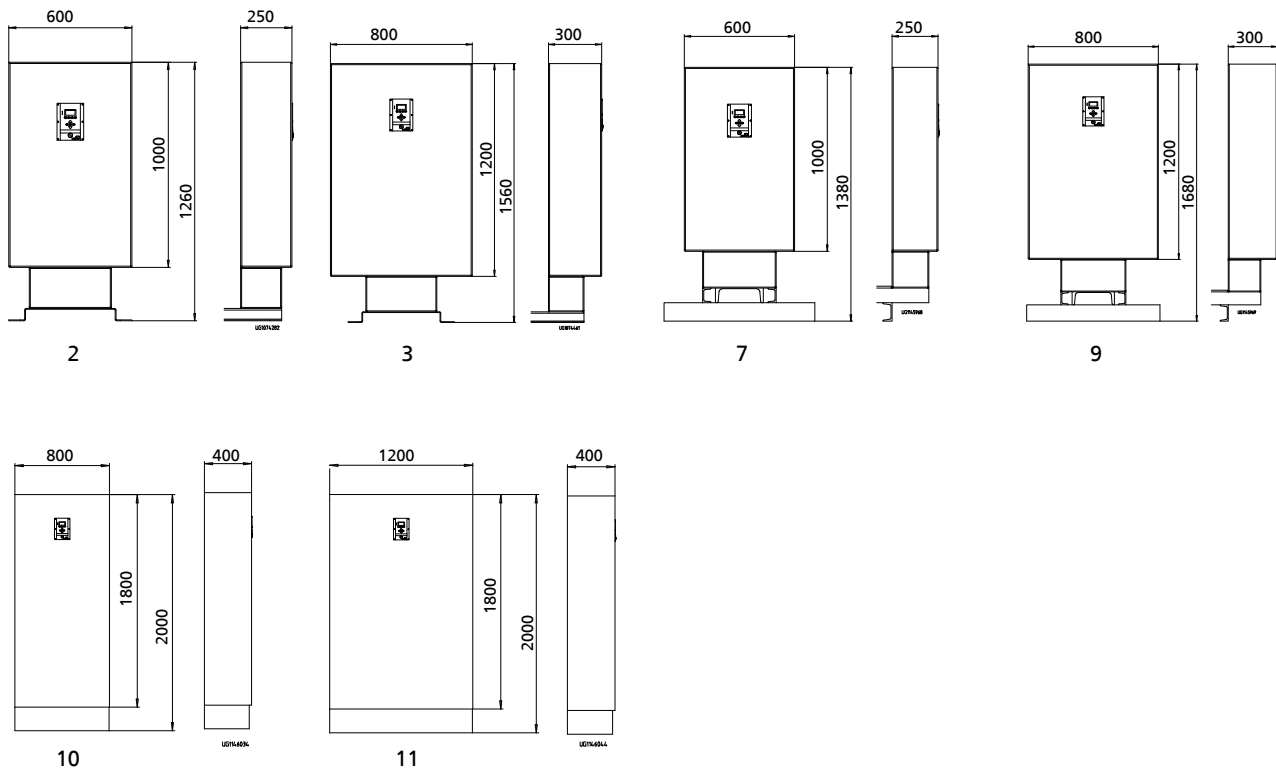
### Control cabinet dimensions

Hyamat V with Movitec 2B, 4B, 6B, 10B and 15B [mm]

| Power per pump | Number of pumps |   |   |    |   |
|----------------|-----------------|---|---|----|---|
| [kW]           | 2               | 3 | 4 | 5  | 6 |
| 4              | 2               |   |   |    |   |
| 5,5            | 3               |   |   | 10 |   |
| 7,5            | 3               |   |   | 10 |   |

Hyamat V with Movitec 25B, 45, 65 and 90B [mm]

| Power per pump | Number of pumps |   |    |    |   |
|----------------|-----------------|---|----|----|---|
| [kW]           | 2               | 3 | 4  | 5  | 6 |
| 4              | 7               |   |    |    |   |
| 5,5            | 9               |   |    | 10 |   |
| 7,5            | 9               |   |    | 10 |   |
| 11             | 9               |   |    | 11 |   |
| 15             | 10              |   |    | 11 |   |
| 18,5           | 10              |   | 11 |    |   |
| 22             | 10              |   | 11 |    |   |



Dimensions of Hyamat V control cabinets

 The control cabinet dimensions refer to systems in standard design. For installation of other optional equipment, larger control cabinets may be required.

## Weights

Weights of the system [kg], weight of the control cabinet ( ) [kg]

| Hyamat V    | 2-2        | 2-1        | 2          | 3-2        | 3-1        | 3          | 4-1        | 4          | 5-1        | 5          | 6-1        | 6          | 7          | 8          | 9   | 10  | 11  | 12  | 13        | 14  | 16  | 18  |
|-------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|-----|-----|-----|-----|-----------|-----|-----|-----|
| 2/B 02../.. | -          | -          | 110        | -          | -          | 111        | -          | 112        | -          | 113        | -          | 117        | 118        | 119        | 123 | 124 | 130 | 131 | -         | 132 | 141 | 143 |
| 3/B 02../.. | -          | -          | 142        | -          | -          | 144        | -          | 145        | -          | 146        | -          | 152        | 154        | 155        | 162 | 163 | 172 | 173 | -         | 176 | 189 | 192 |
| 4/B 02../.. | -          | -          | 172        | -          | -          | 174        | -          | 176        | -          | 177        | -          | 186        | 188        | 189        | 198 | 200 | 212 | 214 | -         | 217 | 234 | 238 |
| 5/B 02../.. | -          | -          | 208        | -          | -          | 210        | -          | 212        | -          | 214        | -          | 226        | 228        | 230        | 241 | 243 | 258 | 260 | -         | 264 | 286 | 290 |
| 6/B 02../.. | -          | -          | 240        | -          | -          | 242        | -          | 244        | -          | 247        | -          | 260        | 262        | 265        | 279 | 281 | 299 | 301 | -         | 306 | 333 | 338 |
| 2/B 04../.. | -          | -          | 110        | -          | -          | 115        | -          | 116        | -          | 120        | -          | 126        | 127        | 134        | 136 | 137 | 143 | 144 | -         | 146 | 174 | -   |
| 3/B 04../.. | -          | -          | 143        | -          | -          | 149        | -          | 151        | -          | 157        | -          | 166        | 167        | 178        | 181 | 182 | 191 | 193 | -         | 196 | 239 | -   |
| 4/B 04../.. | -          | -          | 173        | -          | -          | 181        | -          | 184        | -          | 192        | -          | 204        | 206        | 220        | 224 | 225 | 237 | 240 | -         | 244 | 301 | -   |
| 5/B 04../.. | -          | -          | 209        | -          | -          | 220        | -          | 223        | -          | 233        | -          | 248        | 250        | 268        | 272 | 274 | 290 | 294 | -         | 298 | 369 | -   |
| 6/B 04../.. | -          | -          | 240        | -          | -          | 253        | -          | 257        | -          | 269        | -          | 287        | 289        | 312        | 316 | 319 | 337 | 342 | -         | 347 | 432 | -   |
| 2/B 06../.. | -          | -          | 112        | -          | -          | 121        | -          | 126        | -          | 127        | -          | 136        | 137        | 144        | 145 | 146 | 165 | 166 | -         | 167 | 188 | -   |
| 3/B 06../.. | -          | -          | 145        | -          | -          | 157        | -          | 166        | -          | 167        | -          | 179        | 181        | 191        | 193 | 194 | 224 | 225 | -         | 227 | 258 | -   |
| 4/B 06../.. | -          | -          | 175        | -          | -          | 191        | -          | 203        | -          | 205        | -          | 221        | 223        | 237        | 239 | 241 | 281 | 282 | -         | 284 | 326 | -   |
| 5/B 06../.. | -          | -          | 210        | -          | -          | 230        | -          | 245        | -          | 248        | -          | 268        | 270        | 288        | 290 | 293 | 342 | 344 | -         | 347 | 400 | -   |
| 6/B 06../.. | -          | -          | 240        | -          | -          | 265        | -          | 282        | -          | 285        | -          | 310        | 313        | 334        | 337 | 340 | 399 | 402 | -         | 405 | 468 | -   |
| 2/B 10../.. | -          | -          | 145        | -          | -          | 151        | -          | 162        | -          | 169        | -          | 171        | 189        | 190        | 203 | 205 | 207 | -   | 339       | -   | -   | -   |
| 3/B 10../.. | -          | -          | 195        | -          | -          | 204        | -          | 220        | -          | 231        | -          | 233        | 260        | 263        | 282 | 285 | 288 | -   | 470       | -   | -   | -   |
| 4/B 10../.. | -          | -          | 242        | -          | -          | 255        | -          | 276        | -          | 290        | -          | 294        | 329        | 333        | 359 | 363 | 366 | -   | 595       | -   | -   | -   |
| 5/B 10../.. | -          | -          | 294        | -          | -          | 310        | -          | 336        | -          | 354        | -          | 358        | 402        | 407        | 439 | 444 | 449 | -   | 593 (190) | -   | -   | -   |
| 6/B 10../.. | -          | -          | 341        | -          | -          | 359        | -          | 391        | -          | 412        | -          | 418        | 471        | 476        | 515 | 521 | 527 | -   | 707 (200) | -   | -   | -   |
| 2/B 15../.. | -          | -          | 186        | -          | -          | 205        | -          | 216        | -          | 344        | -          | 346        | 356        | 361        | -   | -   | -   | -   | -         | -   | -   | -   |
| 3/B 15../.. | -          | -          | 252        | -          | -          | 280        | -          | 298        | -          | 474        | -          | 477        | 492        | 499        | -   | -   | -   | -   | -         | -   | -   | -   |
| 4/B 15../.. | -          | -          | 330        | -          | -          | 367        | -          | 391        | -          | 611        | -          | 615        | 634        | 644        | -   | -   | -   | -   | -         | -   | -   | -   |
| 5/B 15../.. | -          | -          | 399        | -          | -          | 578        | -          | 608        | -          | 741 (190)  | -          | 746 (190)  | 771 (190)  | 783 (190)  | -   | -   | -   | -   | -         | -   | -   | -   |
| 6/B 15../.. | -          | -          | 527        | -          | -          | 704        | -          | 739        | -          | 907 (200)  | -          | 912 (200)  | 942 (200)  | 957 (200)  | -   | -   | -   | -   | -         | -   | -   | -   |
| 2/B 25../.. | -          | -          | 371        | -          | -          | 479        | -          | 493        | -          | 697        | -          | 703 (210)  | 597 (210)  | -          | -   | -   | -   | -   | -         | -   | -   | -   |
| 3/B 25../.. | -          | -          | 516        | -          | -          | 663        | -          | 684        | -          | 945        | -          | 954 (210)  | 865 (210)  | -          | -   | -   | -   | -   | -         | -   | -   | -   |
| 4/B 25../.. | -          | -          | 721        | -          | -          | 902        | -          | 930        | -          | 1273       | -          | 1285 (210) | 1198 (210) | -          | -   | -   | -   | -   | -         | -   | -   | -   |
| 5/B 25../.. | -          | -          | 900        | -          | -          | 985 (190)  | -          | 1020 (190) | -          | 1430 (300) | -          | 1445 (300) | 1805       | -          | -   | -   | -   | -   | -         | -   | -   | -   |
| 6/B 25../.. | -          | -          | 1047       | -          | -          | 1106 (200) | -          | 1148 (200) | -          | 1690 (330) | -          | 1708 (330) | 2155       | -          | -   | -   | -   | -   | -         | -   | -   | -   |
| 2 45../..   | -          | 435        | 443        | -          | 625        | 626        | 630        | 528 (210)  | 533 (210)  | 563 (220)  | 568 (220)  | 640 (220)  | -          | -          | -   | -   | -   | -   | -         | -   | -   | -   |
| 3 45../..   | -          | 646        | 658        | -          | 887        | 887        | 894        | 811 (210)  | 818 (210)  | 864 (230)  | 871 (230)  | 978 (230)  | -          | -          | -   | -   | -   | -   | -         | -   | -   | -   |
| 4 45../..   | -          | 818        | 835        | -          | 1135       | 1135       | 1144       | 1065 (210) | 1075 (210) | 1135 (330) | 1145 (330) | 1288 (450) | -          | -          | -   | -   | -   | -   | -         | -   | -   | -   |
| 5 45../..   | -          | 980 (190)  | 1000 (190) | -          | 1356 (300) | 1357 (300) | 1368 (300) | 1738       | 1750       | 1526 (330) | 1538 (330) | 1717 (450) | -          | -          | -   | -   | -   | -   | -         | -   | -   | -   |
| 6 45../..   | -          | 1145 (200) | 1169 (200) | -          | 1597 (330) | 1598 (330) | 1612 (330) | 2071       | 2085       | 1800 (380) | 1814 (380) | 2030 (490) | -          | -          | -   | -   | -   | -   | -         | -   | -   | -   |
| 2 65../..   | -          | -          | 505        | -          | -          | 560        | -          | 699        | -          | 604 (210)  | -          | 611 (210)  | 647 (220)  | 726 (220)  | -   | -   | -   | -   | -         | -   | -   | -   |
| 3 65../..   | -          | -          | 668        | -          | -          | 705        | -          | 914        | -          | 841 (210)  | -          | 851 (230)  | 906 (230)  | 1024 (230) | -   | -   | -   | -   | -         | -   | -   | -   |
| 4 65../..   | -          | -          | 951        | -          | -          | 996        | -          | 1274       | -          | 1209 (210) | -          | 1222 (210) | 1296 (330) | 1453 (450) | -   | -   | -   | -   | -         | -   | -   | -   |
| 5 65../..   | -          | -          | 1060 (190) | -          | -          | 1098 (300) | -          | 1446 (300) | -          | 1832       | -          | 1850       | 1642 (330) | 1838 (450) | -   | -   | -   | -   | -         | -   | -   | -   |
| 6 65../..   | -          | -          | 1240 (200) | -          | -          | 1284 (330) | -          | 1702 (330) | -          | 2181       | -          | 2202       | 1937 (380) | 2174 (490) | -   | -   | -   | -   | -         | -   | -   | -   |
| 2/B 90../.. | 820        | 702 (210)  | 702 (210)  | 756 (220)  | 828 (220)  | 828 (220)  | -          | -          | -          | -          | -          | -          | -          | -          | -   | -   | -   | -   | -         | -   | -   | -   |
| 3/B 90../.. | 1143       | 1036 (210) | 1036 (210) | 1117 (230) | 1203 (230) | 1203 (230) | -          | -          | -          | -          | -          | -          | -          | -          | -   | -   | -   | -   | -         | -   | -   | -   |
| 4/B 90../.. | 1516       | 1405 (210) | 1405 (210) | 1513 (230) | 1657 (450) | 1657 (450) | -          | -          | -          | -          | -          | -          | -          | -          | -   | -   | -   | -   | -         | -   | -   | -   |
| 5/B 90../.. | 1884 (300) | 2214       | 2114       | 2049 (330) | 2229 (450) | 2229 (450) | -          | -          | -          | -          | -          | -          | -          | -          | -   | -   | -   | -   | -         | -   | -   | -   |
| 6/B 90../.. | 2221 (330) | 2362       | 2632       | 2419 (380) | 2635 (490) | 2635 (490) | -          | -          | -          | -          | -          | -          | -          | -          | -   | -   | -   | -   | -         | -   | -   | -   |

**Scope of supply**

Depending on the model, the following items are included in the scope of supply:

**Pressure booster system**

- Two to six vertical high-pressure centrifugal pumps (standard pumps)  
Movitec 2B, 4B, 6B, 10B and 15B with oval flange/round flange  
Movitec 25B, 45, 65 and 90B with round flange
- Hydraulic components made of stainless steel / brass
- One check valve and several shut-off valves for each pump
- Membrane-type accumulator on the discharge side, approved for drinking water
- Pressure transmitter on the discharge side
- Pressure indication via pressure gauge
- Powder-coated/epoxy resin-coated steel baseplate
- Pumps (sizes 2B, 4B, 6B, 10B and 15B) fixed to the baseplate on anti-vibration mounts
- Pressure booster system with pump sizes 25B, 45, 65 and 90B supplied with level-adjustable feet and rubber pads (supplied but not fitted)

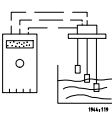
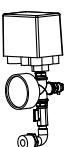
**Control cabinet**

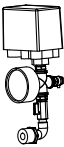
- Control cabinet IP 54
- Pump control and monitoring unit
- Graphical display with operating panel
- LEDs indicating operational availability and fault of the pressure booster system
- Service interface for connection to a PC
- Transformer for control voltage
- Motor protection switch per pump
- Lockable master switch (repair switch)
- Terminals with identification for all connections
- Circuit diagram and list of electric components
- Connection for analog or digital dry running protection equipment
- External ON/OFF connection

## Accessories

 Supplementary equipment / options (E No.) supplied fitted to the pump set  
Accessories (Mat. No.) are included in the delivery but not fitted

### Dry running protection

| Part No.                                                                            | Description                                                                                                                                                                            | Supplementary equipment/<br>option |                         | [kg] | Hyamat K | Hyamat V | Hyamat VP/SVP |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-------------------------|------|----------|----------|---------------|
|                                                                                     |                                                                                                                                                                                        | Factory-fitted                     | Supplied but not fitted |      |          |          |               |
|                                                                                     |                                                                                                                                                                                        | Mat. No.                           | Mat. No.                |      |          |          |               |
|    | Dry running protection float switch<br>With weight, PG cable gland                                                                                                                     |                                    |                         |      |          |          |               |
|                                                                                     | Cable length:                                                                                                                                                                          | 5 m                                | 19071650                | 1    | X        | X        | X             |
|                                                                                     |                                                                                                                                                                                        | 10 m                               | 19070395                | 1.5  | X        | X        | X             |
|                                                                                     |                                                                                                                                                                                        | 20 m                               | 19071651                | 1.8  | X        | X        | X             |
| Special lengths on request                                                          |                                                                                                                                                                                        |                                    |                         |      |          |          |               |
|    | Float switch as dry running protection, with connection cable type H07 RN-F 3x1 mm <sup>2</sup> for lack-of-water monitoring in combination with a reservoir provided by the operator. |                                    |                         |      |          |          |               |
|                                                                                     | Cable length:                                                                                                                                                                          | 5 m                                | 11037630                | 1.1  | X        | X        | X             |
|                                                                                     |                                                                                                                                                                                        | 10 m                               | 11037631                | 1.5  | X        | X        | X             |
|                                                                                     |                                                                                                                                                                                        | 20 m                               | 11037632                | 2    | X        | X        | X             |
| Scope of supply: float switch with connection cable                                 |                                                                                                                                                                                        |                                    |                         |      |          |          |               |
|  | Float switch as dry running protection, with connection cable type H07RN-F 3x1 mm <sup>2</sup> for lack-of-water monitoring in combination with a reservoir provided by the operator.  |                                    |                         |      |          |          |               |
|                                                                                     | Cable length:                                                                                                                                                                          | 3 m                                | 11037742                | 0.5  | X        | X        | X             |
|                                                                                     |                                                                                                                                                                                        | 5 m                                | 11037743                | 0.8  | X        | X        | X             |
|                                                                                     |                                                                                                                                                                                        | 10 m                               | 11037744                | 1.3  | X        | X        | X             |
|                                                                                     |                                                                                                                                                                                        | 15 m                               | 11037745                | 1.8  | X        | X        | X             |
|                                                                                     |                                                                                                                                                                                        | 20 m                               | 11037746                | 2.4  | X        | X        | X             |
|                                                                                     |                                                                                                                                                                                        | 25 m                               | 11037747                | 2.9  | X        | X        | X             |
|                                                                                     |                                                                                                                                                                                        | 30 m                               | 11037748                | 3.4  | X        | X        | X             |
| Scope of supply: float switch with connection cable                                 |                                                                                                                                                                                        |                                    |                         |      |          |          |               |
|  | Weight for adjusting the float switch level                                                                                                                                            |                                    | 18040615                | 1.2  | X        | X        | X             |
|                                                                                     | Scope of supply:<br>Weight with fastening elements                                                                                                                                     |                                    |                         |      |          |          |               |
|  | Set of electrodes as dry running protection, with analysis<br>Cable length to sensor 1.5 m supplied fitted, without connection to control cabinet                                      |                                    | (E420)<br>19075400      | 1.2  | X        | X        | X             |
|                                                                                     | Relay<br>For retrofitting check if the required electrode analysing relay will fit into the control unit (required space: 1 HE)                                                        |                                    | 01069615                | 0.3  | X        | X        | X             |
|                                                                                     | Sensor including terminal box with elements for mounting to inlet tank<br>(Signal cable not included in the scope of supply)                                                           |                                    | 00533947                | 0.9  | X        | X        | X             |
|                                                                                     |                                                                                                                                                                                        |                                    |                         |      |          |          |               |
|  | Dry running protection<br>(Inlet pressure > 1 bar) via pressure switch, wetted part (membrane) made of neoprene                                                                        |                                    |                         |      |          |          |               |
|                                                                                     | For systems with Movitec 2B, 4B, 6B, 10B, 15B                                                                                                                                          |                                    | (E400)<br>19075401      | 1.2  | X        | X        | X             |
|                                                                                     | For systems with Movitec 25B, 45, 65, 90B                                                                                                                                              |                                    | (E401)<br>19075402      | 1.2  | X        | X        | X             |
|                                                                                     | For systems with Movitec 2B, 4B, 6B, 10B, 15B                                                                                                                                          |                                    | 18041026                | 1.2  | X        | X        | X             |
|                                                                                     | For systems with Movitec 25B, 45, 65, 90B                                                                                                                                              |                                    | 18040613                | 1.2  | X        | X        | X             |

| Part No.                                                                                   | Description                                                                                                                                                                                                                                                                                                         | Supplementary equipment/<br>option |                         | [kg] | Hyamat K | Hyamat V | Hyamat VP/SVP |
|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-------------------------|------|----------|----------|---------------|
|                                                                                            |                                                                                                                                                                                                                                                                                                                     | Factory-fitted                     | Supplied but not fitted |      |          |          |               |
|                                                                                            |                                                                                                                                                                                                                                                                                                                     | Mat. No.                           | Mat. No.                |      |          |          |               |
|                                                                                            | Scope of supply:<br>Pressure switch 0 - 10 bar<br>Pressure gauge 0 - 10 bar<br>Tee, hexagon nipple<br>Circuit diagram                                                                                                                                                                                               |                                    |                         |      |          |          |               |
| 693<br>   | Dry running protection<br>(Inlet pressure 0 - 0.5 bar) via pressure transmitter, wetted parts made of 1.4301                                                                                                                                                                                                        |                                    |                         |      |          |          |               |
|                                                                                            | For systems with Movitec 2B, 4B, 6B, 10B, 15B                                                                                                                                                                                                                                                                       | (E461)<br>19075403                 |                         | 1.2  | X        | X        | X             |
|                                                                                            | For systems with Movitec 25B, 45, 65, 90B                                                                                                                                                                                                                                                                           | (E460)<br>19075404                 |                         | 1.2  | X        | X        | X             |
|                                                                                            | For systems with Movitec 2B, 4B, 6B, 10B, 15B                                                                                                                                                                                                                                                                       |                                    | 18041182                | 1.2  | X        | X        | X             |
|                                                                                            | For systems with Movitec 25B, 45, 65, 90B                                                                                                                                                                                                                                                                           |                                    | 18041183                | 1.2  | X        | X        | X             |
|                                                                                            | Scope of supply:<br>Pressure transmitter 0 - 1 bar, 4 - 20mA<br>Pressure gauge -1 - 1.5 bar<br>Tee, hexagon nipple<br>Circuit diagram                                                                                                                                                                               |                                    |                         |      |          |          |               |
| 693<br>  | Dry running protection<br>(Inlet pressure > 0.5 bar) via pressure transmitter, wetted parts made of 1.4301                                                                                                                                                                                                          |                                    |                         |      |          |          |               |
|                                                                                            | For systems with Movitec 2B, 4B, 6B, 10B, 15B                                                                                                                                                                                                                                                                       | (E410)<br>19075405                 |                         | 1.2  | X        | X        | X             |
|                                                                                            | For systems with Movitec 25B, 45, 65, 90B                                                                                                                                                                                                                                                                           | (E411)<br>19075406                 |                         | 1.2  | X        | X        | X             |
|                                                                                            | For systems with Movitec 2B, 4B, 6B, 10B, 15B                                                                                                                                                                                                                                                                       |                                    | 18041027                | 1.5  | X        | X        | X             |
|                                                                                            | For systems with Movitec 25B, 45, 65, 90B                                                                                                                                                                                                                                                                           |                                    | 18040614                | 1.5  | X        | X        | X             |
|                                                                                            | Scope of supply:<br>Pressure transmitter 0 - 10 bar, 4 - 20mA<br>Pressure gauge 0 - 10 bar<br>Tee, hexagon nipple<br>Circuit diagram                                                                                                                                                                                |                                    |                         |      |          |          |               |
| 82-16                                                                                      | Dry running protection for suction lift operation (flow monitoring)<br>The system stops when no flow is detected on the suction side and at the same time the discharge-side pressure drops below an adjustable value.<br><b>Note:</b><br>Automatic reset is not possible with this type of dry running protection! | (E841)<br>19075476                 |                         | 0.3  | X        | X        | X             |
| 693<br> | Dry running protection for Hyamat K/V/VP made of stainless steel C1 or C2<br>(Inlet pressure > 1 bar) via pressure switch, wetted part (membrane) made of neoprene                                                                                                                                                  |                                    |                         |      |          |          |               |
|                                                                                            | For systems with Movitec 2B, 4B, 6B, 10B, 15B                                                                                                                                                                                                                                                                       | (E430)<br>19075407                 |                         | 1.2  | X        | X        | X             |
|                                                                                            | For systems with Movitec 25B, 45, 65, 90B                                                                                                                                                                                                                                                                           | (E431)<br>19075408                 |                         | 1.2  | X        | X        | X             |
|                                                                                            | For systems with Movitec 2B, 4B, 6B, 10B, 15B                                                                                                                                                                                                                                                                       |                                    | 18041159                | 1.2  | X        | X        | X             |
|                                                                                            | For systems with Movitec 25B, 45, 65, 90B                                                                                                                                                                                                                                                                           |                                    | 18041160                | 1.2  | X        | X        | X             |
|                                                                                            | Scope of supply:<br>Pressure switch 0 - 10 bar<br>Pressure gauge 0 - 10 bar<br>Tee, hexagon nipple<br>Circuit diagram                                                                                                                                                                                               |                                    |                         |      |          |          |               |
| 693                                                                                        | Dry running protection for Hyamat K/V/VP made of stainless steel C1 or C2<br>(Inlet pressure > 1 bar) via pressure transmitter, wetted part made of V4A                                                                                                                                                             |                                    |                         |      |          |          |               |

| Part No.                                                                          | Description                                                                                                                          | Supplementary equipment/<br>option |                         | [kg] | Hyamat K | Hyamat V | Hyamat VP/SVP |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-------------------------|------|----------|----------|---------------|
|                                                                                   |                                                                                                                                      | Factory-fitted                     | Supplied but not fitted |      |          |          |               |
|                                                                                   |                                                                                                                                      | Mat. No.                           | Mat. No.                |      |          |          |               |
|  | For systems with Movitec 2B, 4B, 6B, 10B, 15B                                                                                        | (E432)<br>19075409                 |                         | 1.2  | X        | X        | X             |
|                                                                                   | For systems with Movitec 25B, 45, 65, 90B                                                                                            | (E433)<br>19075410                 |                         | 1.2  | X        | X        | X             |
|                                                                                   | For systems with Movitec 2B, 4B, 6B, 10B, 15B                                                                                        |                                    | 18041157                | 2    | X        | X        | X             |
|                                                                                   | For systems with Movitec 25B, 45, 65, 90B                                                                                            |                                    | 18041158                | 2    | X        | X        | X             |
|                                                                                   | Scope of supply:<br>Pressure transmitter 0 - 10 bar, 4 - 20mA<br>Pressure gauge 0 - 10 bar<br>Tee, hexagon nipple<br>Circuit diagram |                                    |                         |      |          |          |               |

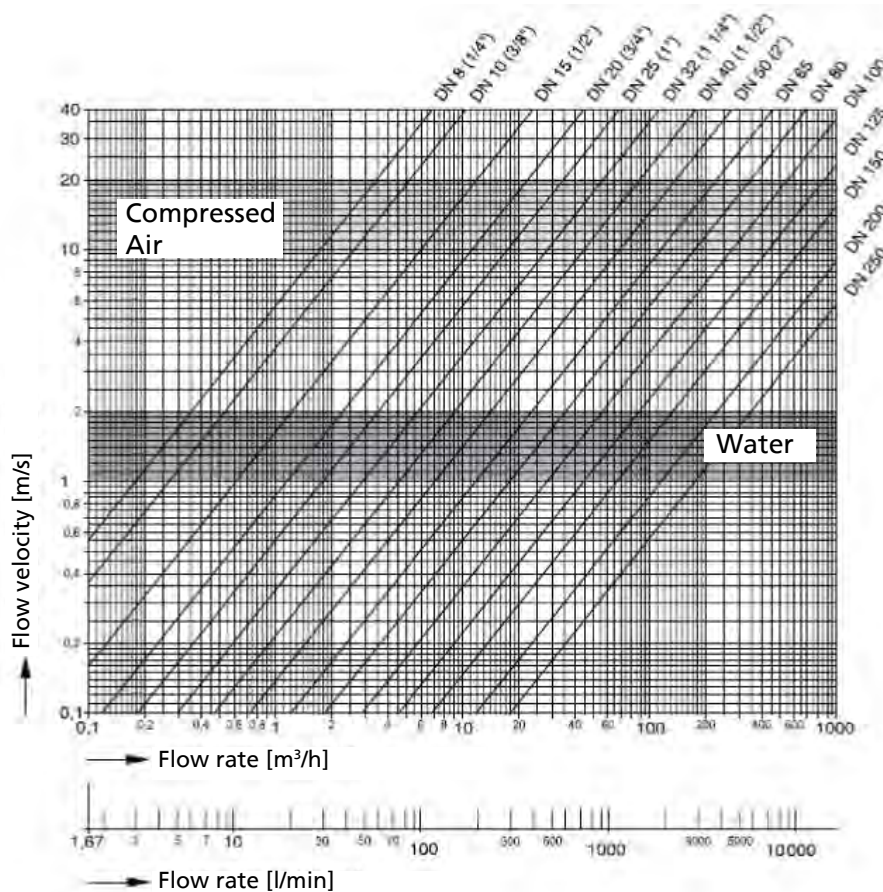
### Pressure reducer

Pressure reducer

| Part No. | Size    | Type Honeywell | Max. inlet pressure | Downstream pressure | Mat. No. | [kg] | Hyamat K | Hyamat V | Hyamat VP/SVP |
|----------|---------|----------------|---------------------|---------------------|----------|------|----------|----------|---------------|
|          |         |                | [bar]               | [bar]               |          |      |          |          |               |
| 69-7     | R 1     | D06 F...A      | 25                  | 1,5 - 6             | 01057183 | 1.4  | X        | X        | X             |
| 69-7     | R 1     | D06 FN...B     | 25                  | 0,5 - 2             | 01057184 | 2.4  | X        | X        | X             |
| 69-7     | R 1 1/4 | D06 F...A      | 25                  | 1,5 - 6             | 01057185 | 2    | X        | X        | X             |
| 69-7     | R 1 1/4 | D06 FN...B     | 25                  | 0,5 - 2             | 01057186 | 2.8  | X        | X        | X             |
| 69-7     | R 2     | D06 F...A      | 25                  | 1,5 - 6             | 00522969 | 4.5  | X        | X        | X             |
| 69-7     | R 2     | D06 FN...B     | 25                  | 0,5 - 2             | 00522960 | 5.6  | X        | X        | X             |
| 69-7     | DN 65   | D15 P          | 16                  | 1,5 - 6             | 00522957 | 28.2 | X        | X        | X             |
| 69-7     | DN 65   | D15 NP         | 16                  | 0,2 - 2             | 00522959 | 37   | X        | X        | X             |
| 69-7     | DN 80   | D15 P          | 16                  | 1,5 - 6             | 00522958 | 41.5 | X        | X        | X             |
| 69-7     | DN 80   | D15 NP         | 16                  | 0,2 - 2             | 00522956 | 54   | X        | X        | X             |
| 69-7     | DN 100  | D15 P          | 16                  | 1,5 - 6             | 00119996 | 67   | X        | X        | X             |
| 69-7     | DN 100  | D15 NP         | 16                  | 0,2 - 2             | 00119997 | 87.5 | X        | X        | X             |
| 69-7     | DN 150  | D15 P          | 16                  | 1,5 - 6             | 00198244 | 150  | X        | X        | X             |
| 69-7     | DN 150  | D15 NP         | 16                  | 0,2 - 2             | 00198247 | 196  | X        | X        | X             |
| 69-7     | DN 200  | D15 P          | 16                  | 1,5 - 6             | 00198245 | 408  | X        | X        | X             |
| 69-7     | DN 200  | D15 NP         | 16                  | 0,2 - 2             | 00198248 | 580  | X        | X        | X             |

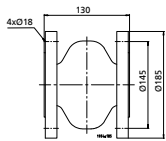
Flanges drilled to PN 16, DIN 2533

### Selection diagram for pressure reducers



### Expansion joints for vibration damping

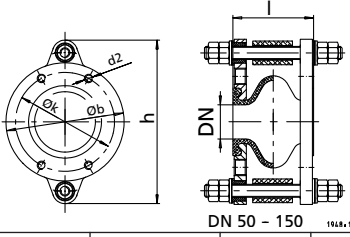
Expansion joints PN 10

| Part No.                                                                            | Description                                                          | Mat. No.         | [kg]     | Hyamat K | Hyamat V | Hyamat VP/SVP |   |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------|------------------|----------|----------|----------|---------------|---|
| Connection option type 50                                                           |                                                                      |                  |          |          |          |               |   |
| Expansion joints PN 10                                                              |                                                                      |                  |          |          |          |               |   |
|  | DN 40 with adapter flange PN 10/16 type 50 with KTW approval         | 01125069         | 3.4      | X        | X        | X             |   |
|                                                                                     | DN 65 with adapter flange PN 10/16 type 50 with KTW approval         | 11037184         | 5.6      | X        | X        | X             |   |
|                                                                                     | DN 80 with adapter flange PN 10/16 type 50 with KTW approval         | 01125072         | 6.7      | X        | X        | X             |   |
|                                                                                     |                                                                      |                  |          |          |          |               |   |
|  | Screwed flange R 1½ (internal thread) with screws, bolts and washers | Mat. galv. steel | 18040968 | 5        | X        | X             | X |
|                                                                                     | For expansion joint DN 40, type 49                                   | Mat. 1.4541      | 18040969 | 5        | X        | X             | X |
|                                                                                     | Screwed flange R 2 (internal thread) with screws, bolts and washers  | Mat. galv. steel | 18040647 | 5        | X        | X             | X |
|                                                                                     | For expansion joint DN 50, type 50                                   | Mat. 1.4541      | 18040646 | 5        | X        | X             | X |
|                                                                                     | Screwed flange R 2½ (internal thread) with screws, bolts and washers | Mat. galv. steel | 18040649 | 6        | X        | X             | X |
|                                                                                     | For expansion joint DN 65, 49                                        | Mat. 1.4541      | 18040648 | 7        | X        | X             | X |
|                                                                                     | Screwed flange R 3 (internal thread) with screws, bolts and washers  | Mat. St-TZN      | 18040970 | 7        | X        | X             | X |
|                                                                                     | For expansion joint DN 80, type 49                                   | Mat. 1.4541      | 18040971 | 7        | X        | X             | X |

Expansion joints PN 16

Type 49, blue, Variant C with integrated length limiter

Drinking water-approved as per regulation of the German health authority (Bundesgesundheitsamt) of 1986

| Part No.                                                                          | Size   | b   | h   | l   | DN  | k   | d2      | Material                    | Mat. No. | [kg] | Hyamat K | Hyamat V | Hyamat VP/VP |
|-----------------------------------------------------------------------------------|--------|-----|-----|-----|-----|-----|---------|-----------------------------|----------|------|----------|----------|--------------|
|  |        |     |     |     |     |     |         |                             |          |      |          |          |              |
| 71-8                                                                              | DN 40  | 150 | 250 | 100 | 40  | 110 | 4x M16  | Rubber,<br>galvanised steel | 01125068 | 4.8  | X        | X        | X            |
| 71-8                                                                              | DN 50  | 165 | 265 | 100 | 50  | 125 | 4x M16  |                             | 01057406 | 8    | X        | X        | X            |
| 71-8                                                                              | DN 65  | 185 | 285 | 100 | 65  | 145 | 4x M16  |                             | 01057407 | 8    | X        | X        | X            |
| 71-8                                                                              | DN 80  | 200 | 290 | 100 | 80  | 160 | 8x M16  |                             | 01049847 | 5.5  | X        | X        | X            |
| 71-8                                                                              | DN 100 | 220 | 320 | 100 | 100 | 180 | 8x M16  |                             | 01049848 | 6.6  | X        | X        | X            |
| 71-8                                                                              | DN 150 | 285 | 385 | 100 | 150 | 240 | 8x M20  |                             | 01049850 | 11.4 | X        | X        | X            |
| 71-8                                                                              | DN 200 | 340 | 440 | 100 | 200 | 295 | 12x M20 |                             | 11037185 | 10   | X        | X        | X            |

### Membrane-type accumulator - steel

- Operating pressure up to 16 bar
- Operating temperature 70 °C
- Pressure gauge
- Feed valve
- Drain socket
- Replaceable membrane to DIN 4807, Part 3, suitable for drinking water in accordance with the German KTW recommendations, category C
- Accumulator's inside walls powder-coated in accordance with KTW recommendations, category C
- Service pipe lined with thermoplastic material, suitable for drinking water in accordance with KTW recommendations, category C
- Anti-corrosive exterior coating, green RAL 6018
- Accumulator pre-charged to 4 bar with nitrogen (data not stamped onto name plate)
- DN 80/100 adapter on request

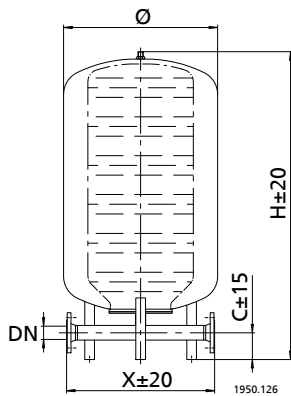


Fig. 1

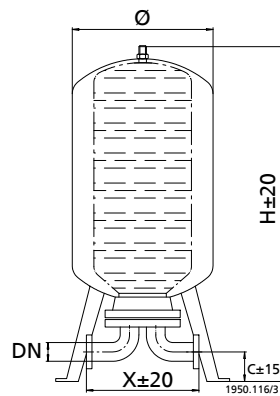


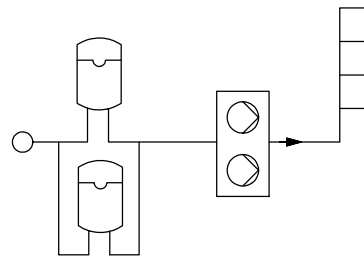
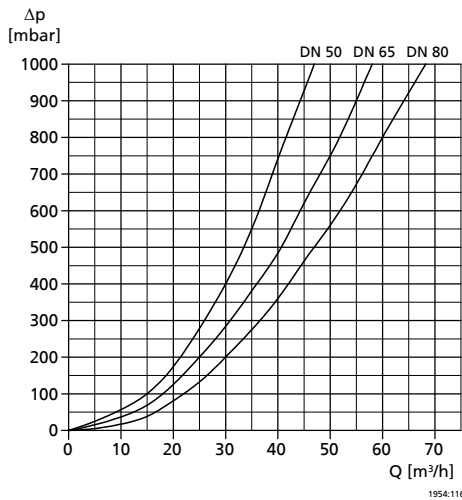
Fig. 2

### Membrane-type accumulator - steel

| Part No. |        | Description                                             |                      |    | Mat. No. | [kg] | Hyamat K | Hyamat V | Hyamat VP/SVP |
|----------|--------|---------------------------------------------------------|----------------------|----|----------|------|----------|----------|---------------|
|          |        | Type                                                    | Effective volume [l] | DN |          |      |          |          |               |
| 591      | Fig. 1 | Membrane-type accumulator with double connection, PN 10 |                      |    |          |      |          |          |               |
|          |        | 80 l                                                    | 60                   | 50 | 01064950 | 33   | X        | X        | X             |
|          |        | 200 l                                                   | 150                  | 50 | 01065001 | 56   | X        | X        | X             |
|          |        | 300 l                                                   | 225                  | 50 | 01065002 | 61   | X        | X        | X             |
|          |        | 500 l                                                   | 350                  | 50 | 01065003 | 91   | X        | X        | X             |
|          |        | 600 l                                                   | 340                  | 50 | 01056394 | 258  | X        | X        | X             |
|          |        | 800 l                                                   | 450                  | 50 | 00198687 | 270  | X        | X        | X             |
|          |        | 1000 l                                                  | 450                  | 50 | 00198688 | 345  | X        | X        | X             |
|          |        | 1001 l                                                  | 750                  | 65 | 01054867 | 420  | X        | X        | X             |
|          | Fig. 2 | Membrane-type accumulator with double connection, PN 16 |                      |    |          |      |          |          |               |
|          |        | 80 l                                                    | 60                   | 50 | 00198689 | 70   | X        | X        | X             |
|          |        | 200 l                                                   | 150                  | 50 | 00198690 | 116  | X        | X        | X             |
|          |        | 300 l                                                   | 225                  | 50 | 00198691 | 110  | X        | X        | X             |
|          |        | 400 l                                                   | 300                  | 50 | 00198692 | 180  | X        | X        | X             |
|          |        | 600 l                                                   | 340                  | 50 | 01056395 | 298  | X        | X        | X             |
|          |        | 800 l                                                   | 450                  | 50 | 00198693 | 270  | X        | X        | X             |
|          |        | 1000 l                                                  | 450                  | 50 | 00198694 | 345  | X        | X        | X             |
|          |        | 1001 l                                                  | 750                  | 65 | 01054868 | 530  | X        | X        | X             |
|          |        | Membrane rupture detector (from accumulator type 600)   |                      |    | 01147508 | 1    | X        | X        | X             |

### Connection of accumulators with double connection for large flow rates

Accumulators with double connection must be used for drinking water applications in accordance with DIN 4807-5. On request, the documented accumulators are available with larger connection cross-sections for larger flow rates.

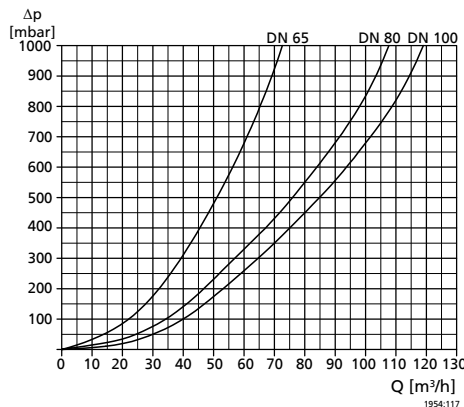


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2x DN 65  $\approx$  DN 100  
2x DN 80  $\approx$  DN 125  
2x DN 100  $\approx$  DN 150

Installation and connection examples for larger flow rates (suction side)

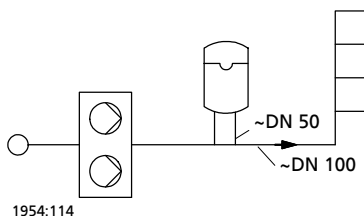
Accumulators 300, 400, 500, 800 and 1000, documented with DN 50



Accumulator 1001, documented with DN 65

### Installation and connection examples

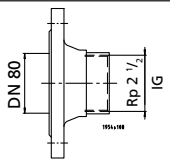
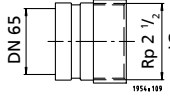
For large flow rates on the **discharge side** smaller accumulators with small connection diameters can be fitted in the bypass line.



1954:114

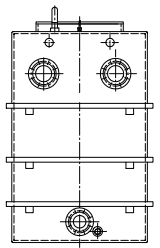
Bypass operation is not possible on the **suction side**. To accommodate larger flow rates several accumulators must be connected in parallel.

## Adapters

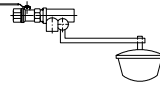
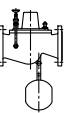
| Part No. |                                                                                   | Description                                                                                             | Mat. No. | [kg] | Hyamat |
|----------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------|------|--------|
| 82-5     |  | Adapter 2 1/2" - DN 80<br>Made of 1.4301<br>For connecting Hyamat to larger pipes                       | 11036865 | 0.4  | X      |
| 82-5     |  | Adapter 2 1/2" - Victaulic coupling DN 65<br>Made of 1.4301<br>For connecting Hyamat to Victaulic pipes | 11036866 | 0.4  | X      |

## Inlet tank with accessories

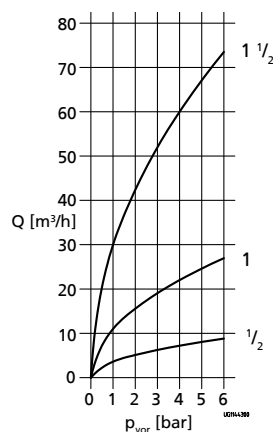
### Inlet tank

| Part No.                                                                            | Total volume [l]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Effective volume [l] | Connection Inlet line | Connection Outlet line | Mat. No. | [kg] |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------|------------------------|----------|------|
|   | <b>Inlet tank with Class A air gap</b> to DIN EN 1717 and DIN EN 13077, under atmospheric pressure, made of polyethylene, generally recognised as safe and therefore food-approved, max. operating temperature 50 °C, for use with indirect connection to public drinking water systems. With inspection cover (diam. = 280 mm up to 540 l, diam. = 600 mm from 800 l of effective volume), drain plug and protective caps.<br>Tank design to DIN EN 1717 and DIN EN 13077<br>Round tanks with effective volumes of up to 540 l, rectangular tanks with effective volumes of 800 l and above, front face overflow.<br>Requires combination with inlet and dry running protection sets. |                      |                       |                        |          |      |
| 591.01                                                                              | 300                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 150                  | R 1                   | DN 50                  | 01141899 | 30   |
| 591.01                                                                              | 600                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 300                  | R 1 1/2               | DN 50                  | 01141900 | 47   |
| 591.01                                                                              | 750                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 540                  | R 1 1/2               | DN 50                  | 01141901 | 50   |
| 591.01                                                                              | 1500                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 800                  | 2 x R 2               | DN 80                  | 01371575 | 157  |
| 591.01                                                                              | 850                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 600                  | R 1 1/2               | DN 50                  | 01374941 | 56   |
| 591.01                                                                              | 1800                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1100                 | 2 x R 2               | DN 80                  | 01371576 | 170  |
| 591.01                                                                              | 2700                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1500                 | 2 x R 2               | DN 100                 | 01371657 | 250  |
| 591.01                                                                              | 3200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2000                 | 2 x R 2               | DN 125                 | 01371658 | 330  |
| 591.01                                                                              | 2700                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1500                 | DN 80                 | DN 100                 | 01371659 | 250  |
| 591.01                                                                              | 3200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2000                 | DN 100                | DN 125                 | 01371660 | 330  |
|  | <b>Closed inlet tank made of polyethylene, under atmospheric pressure,</b> generally recognised as safe and therefore food-approved, max. operating temperature 50 °C, for use with indirect connection to public drinking water systems. With inspection cover (diam. = 280 mm up to 500 l, diam. = 600 mm from 800 l of effective volume), drain plug and protective caps.<br>Tank design to DIN 1988<br>Round tanks with effective volumes of up to 500 l, rectangular tanks with effective volumes of 800 l and above<br>Requires combination with inlet and dry running protection sets.                                                                                          |                      |                       |                        |          |      |
| 591.01                                                                              | 300                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 150                  | R 1                   | DN 50                  | 01136287 | 30   |
| 591.01                                                                              | 600                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 300                  | R 1 1/2               | DN 50                  | 01136288 | 40   |
| 591.01                                                                              | 750                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 500                  | R 1 1/2               | DN 50                  | 01136289 | 50   |
| 591.01                                                                              | 1500                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 800                  | 2 x R 2               | DN 80                  | 01136290 | 150  |
| 591.01                                                                              | 1800                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1000                 | 2 x R 2               | DN 80                  | 01136291 | 170  |
| 591.01                                                                              | 2700                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1500                 | 2 x R 2               | DN 100                 | 01136292 | 250  |
| 591.01                                                                              | 3200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2000                 | 2 x R 2               | DN 125                 | 01136293 | 330  |
| 591.01                                                                              | 2700                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1500                 | DN 80                 | DN 100                 | 01136294 | 250  |
| 591.01                                                                              | 3200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2000                 | DN 100                | DN 125                 | 01136295 | 330  |

### Accessories

| Part No.                                                                                           | Description                                                                                                                                                                                                                                                                                                        | Connection | Mat. No. | [kg] |
|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|------|
| 81-42 and 741<br> | Float valve inlet set for drinking water installations<br>Scope of supply: float valve, ball valve and gasket, approved for drinking water applications                                                                                                                                                            | R 1        | 19070392 | 1.5  |
|                                                                                                    |                                                                                                                                                                                                                                                                                                                    | R 1½       | 19070393 | 2.5  |
|                                                                                                    |                                                                                                                                                                                                                                                                                                                    | R 2        | 19070394 | 3.2  |
| 741<br>           | Diaphragm valve inlet set<br>Scope of supply: float valve and pilot valve for control function, minimum inlet pressure: 0.8 bar (for PE inlet tanks with 1500 or 2000 l effective volume) with installation drawing<br>For inlet pressures exceeding 5 bar a pressure reducer must be installed in the inlet line. | DN 80      | 19071381 | 30   |
|                                                                                                    |                                                                                                                                                                                                                                                                                                                    | DN 100     | 19071382 | 40   |
| 81-42<br>         | BOA-Compact EKB shut-off valve<br>DVGW                                                                                                                                                                                                                                                                             | DN 80      | 19071383 | 18   |
|                                                                                                    |                                                                                                                                                                                                                                                                                                                    | DN 100     | 19071384 | 21   |
| 70-3<br>          | Solenoid valve, 24 V AC/DC                                                                                                                                                                                                                                                                                         |            |          |      |
|                                                                                                    | Solenoid valve ½" (kv = 3,6 m³/h)                                                                                                                                                                                                                                                                                  |            | 19074196 | 1.4  |
|                                                                                                    | Solenoid valve 1" (kv = 11 m³/h)                                                                                                                                                                                                                                                                                   |            | 19074197 | 1.8  |
|                                                                                                    | Solenoid valve 1½" (kv = 30 m³/h)                                                                                                                                                                                                                                                                                  |            | 19074198 | 2    |
| 70-3<br>         | Flushing device for inlet set<br>Includes solenoid valve and digital timer<br>With connection cable (approx. 2.5 m) and shockproof plug<br>(not suitable for fully desalinated water)                                                                                                                              |            |          |      |
|                                                                                                    | Solenoid valve ½" (kv = 3,6 m³/h)                                                                                                                                                                                                                                                                                  |            | 19074174 | 1.5  |
|                                                                                                    | Solenoid valve 1" (kv = 11 m³/h)                                                                                                                                                                                                                                                                                   |            | 19074175 | 2    |
|                                                                                                    | Solenoid valve 1½" (kv = 30 m³/h)                                                                                                                                                                                                                                                                                  |            | 19074176 | 2.5  |

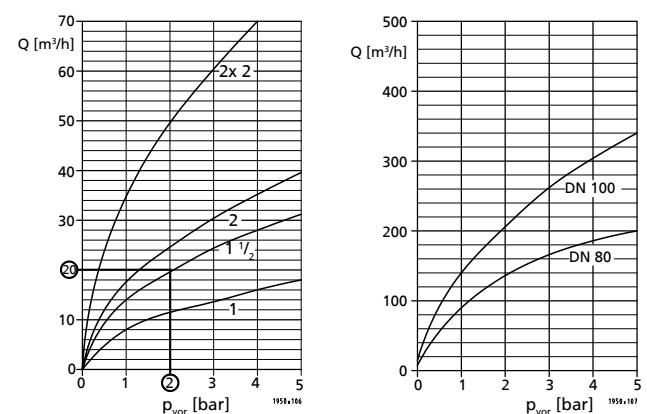
### Selection diagram for solenoid valves



**Example:**  
Inlet pressure  $p_{in} = 2$  bar  
Flow rate  $Q = 35$  m³/h

**Selected:** solenoid valve 1 1/2

### Selection diagram for selecting inlet valves

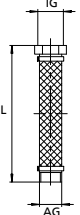


**Example:**  
Flow rate  $Q = 20$  m³/h  
Inlet pressure  $p_{in} = 2$  bar

**Result:** float valve 1 1/2

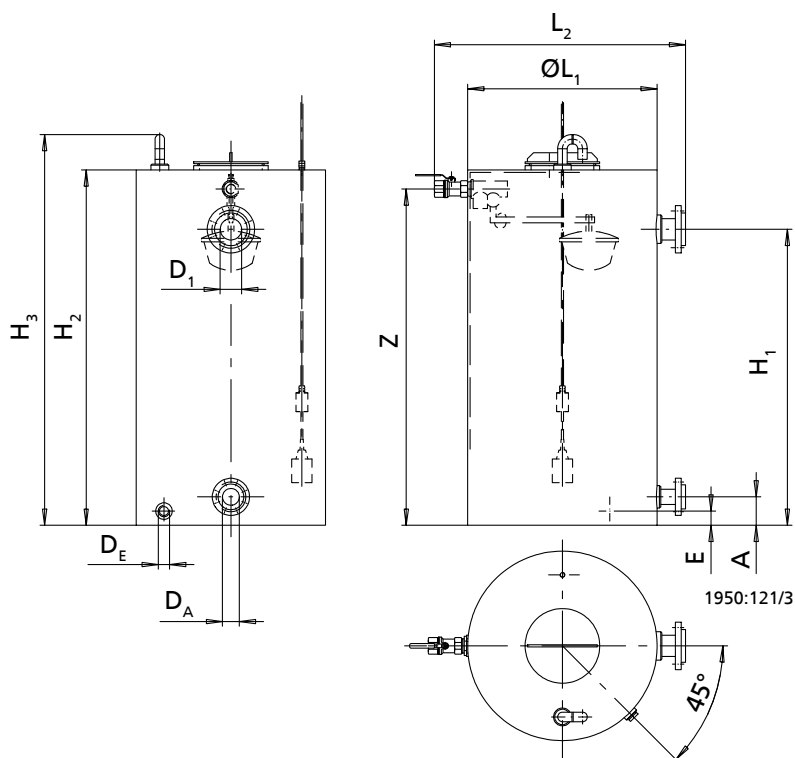
Suitable tank: total volume 600 l (without volume compensation)

### Material required for installation

| Part No.                                                                          | Description                                                            | Supplementary equipment/<br>option | Mat. No. | [kg] | Hyamat K | Hyamat V | Hyamat VP/SVP |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------|----------|------|----------|----------|---------------|
|  | Flexible hose PN 10, R 1, L = 300 mm (without drinking water approval) |                                    | 11037177 | 2.4  | X        | X        | X             |
|                                                                                   | Flexible hose PN 10, R 1 ¼, L = 300 mm (with drinking water approval)  |                                    | 11037178 | 2.5  | X        | X        | X             |
|                                                                                   | Flexible hose PN 10, R 1 ½, L = 350 mm (with drinking water approval)  |                                    | 11037179 | 3    | X        | X        | X             |
|                                                                                   | To compensate for installation offsets or flexibly connect the system. |                                    |          |      |          |          |               |

### Dimensions of the PE inlet tank

#### Dimensions of the PE inlet tank to DIN 1988 [mm]



Effective volume of 150 - 500 litres

Connection flanges to DIN 2642, drilled to DIN 2533

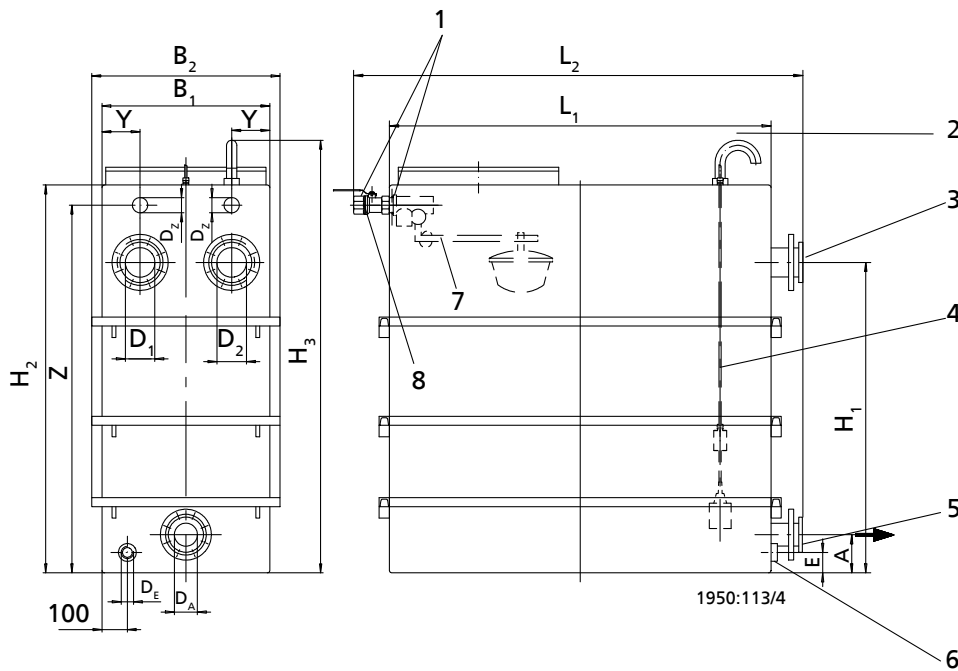


Fig. A

Effective volume of 800, 1000, 1500, 2000 litres

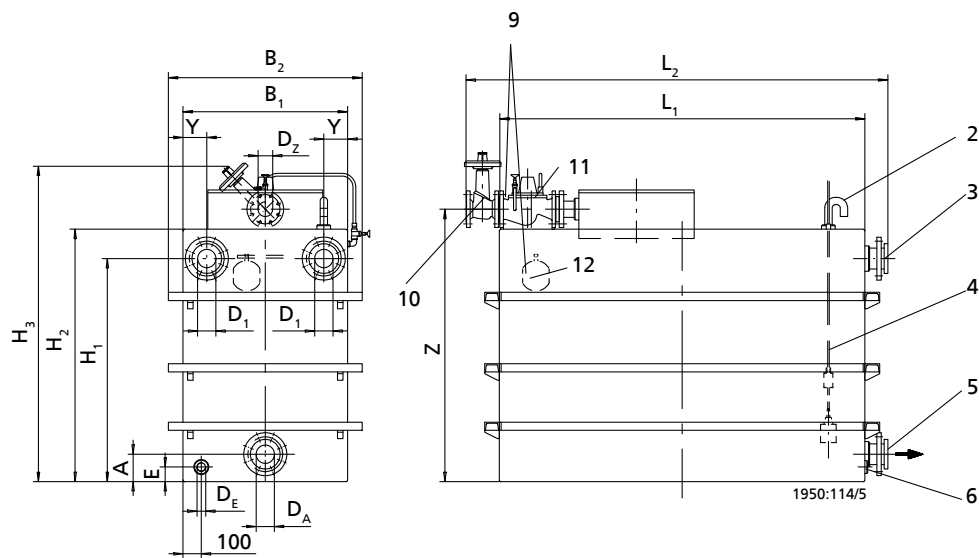


Fig. B

Effective volume of 1500 and 2000 litres

|    |                                       |    |                                                   |
|----|---------------------------------------|----|---------------------------------------------------|
| 1  | Float valve inlet set                 | 2  | Opening for venting and aeration with fine screen |
| 3  | Overflow                              | 4  | Dry running protection set (accessory)            |
| 5  | Outlet to the pressure booster system | 6  | Drain                                             |
| 7  | Float valve                           | 8  | Ball valve                                        |
| 9  | Diaphragm valve inlet set (accessory) | 10 | Shut-off valve (accessory)                        |
| 11 | Diaphragm valve                       | 12 | Pilot valve for diaphragm valve                   |

### Dimensions of the PE inlet tank to DIN 1988 [mm]

Dimensions [mm]

| Effective volume | Total volume | L <sub>1</sub> | L <sub>2</sub> <sup>4)</sup> | B <sub>1</sub> | B <sub>2</sub> | H <sub>1</sub> | H <sub>2</sub> | H <sub>3</sub> | D <sub>1</sub> | D <sub>2</sub> | D <sub>A</sub> | D <sub>E</sub> | Z    | A    | E    | Y    | Fig. |
|------------------|--------------|----------------|------------------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|------|------|------|------|------|
| [l]              | [l]          | [mm]           | [mm]                         | [mm]           | [mm]           | [mm]           | [mm]           | [mm]           |                |                |                |                | [mm] | [mm] | [mm] | [mm] |      |
| 150              | 300          | Ø640           | 855                          | Ø640           | -              | 790            | 1000           | 1150           | G 2            | G 1            | DN 50          | G 1            | 920  | 120  | 60   | 100  | A    |
| 300              | 600          | Ø800           | 1045                         | Ø800           | -              | 1000           | 1250           | 1400           | DN 80          | G 1 1/2        | DN 50          | G 1            | 1170 | 120  | 60   | 100  | A    |
| 500              | 750          | Ø800           | 1045                         | Ø800           | -              | 1250           | 1500           | 1650           | DN 80          | G 1 1/2        | DN 50          | G 1            | 1420 | 120  | 60   | 100  | A    |
| 800              | 1500         | 1500           | 1865                         | 660            | 750            | 1220           | 1524           | 1700           | 2x DN 100      | 2x G 2         | DN 80          | G 1 1/2        | 1445 | 150  | 80   | 130  | A    |
| 1000             | 1800         | 1850           | 2215                         | 660            | 800            | 1220           | 1524           | 1700           | 2x DN 100      | 2x G 2         | DN 80          | G 1 1/2        | 1445 | 150  | 80   | 130  | A    |
| 1500             | 2700         | 2000           | 2365                         | 900            | 1060           | 1220           | 1524           | 1700           | 2x DN 100      | 2x G 2         | DN 100         | G 1 1/2        | 1445 | 150  | 80   | 130  | A    |
| 2000             | 3200         | 2000           | 2365                         | 900            | 1060           | 1490           | 1800           | 1980           | 2x DN 100      | 2x G 2         | DN 125         | G 1 1/2        | 1720 | 150  | 80   | 130  | A    |
| 1500             | 2700         | 2000           | 2495                         | 900            | 1080           | 1220           | 1380           | 1800           | 2x DN 100      | DN 80          | DN 100         | G 1 1/2        | 1490 | 150  | 80   | 130  | B    |
| 2000             | 3200         | 2000           | 2525                         | 900            | 1080           | 1490           | 1650           | 2200           | 2x DN 100      | DN 100         | DN 125         | G 1 1/2        | 1770 | 150  | 80   | 130  | B    |

### Dimensions of the PE inlet tank with air gap to DIN EN 1717/DIN EN 13077

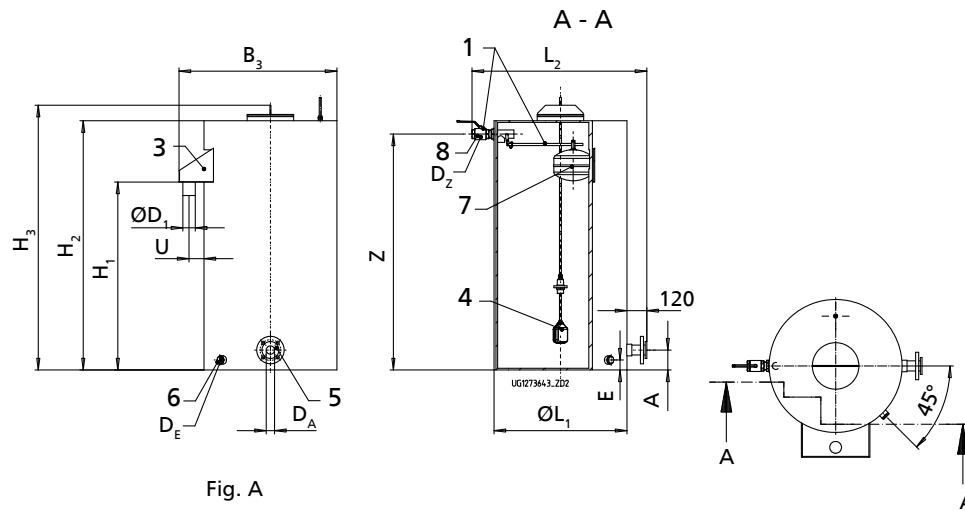


Fig. A

Effective volume of 150 to 600 litres

4) Shut-off valve inlet set included

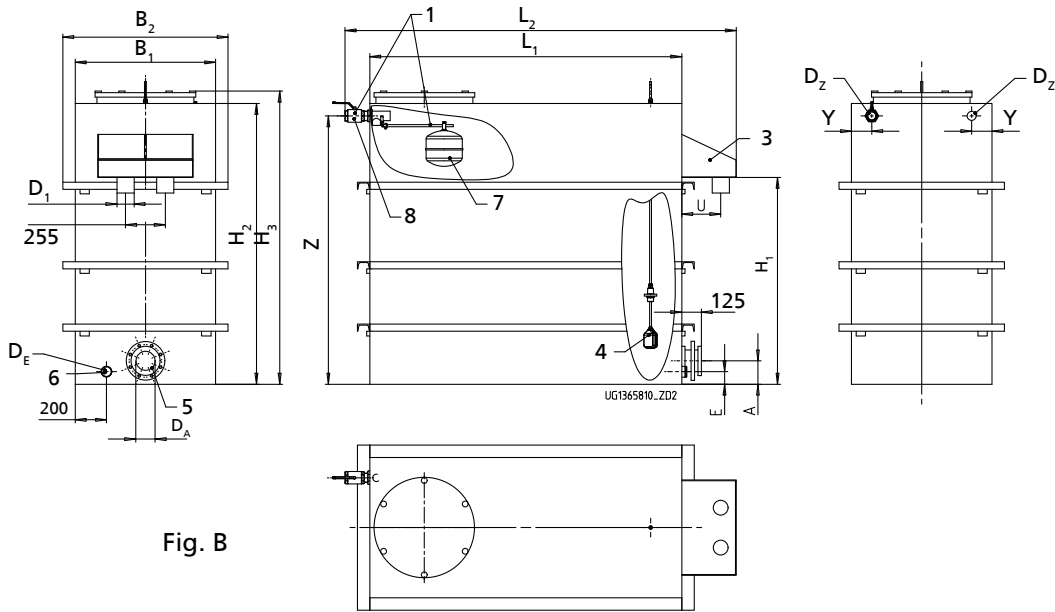


Fig. B

Effective volume of 800, 1000, 1500, 2000 litres

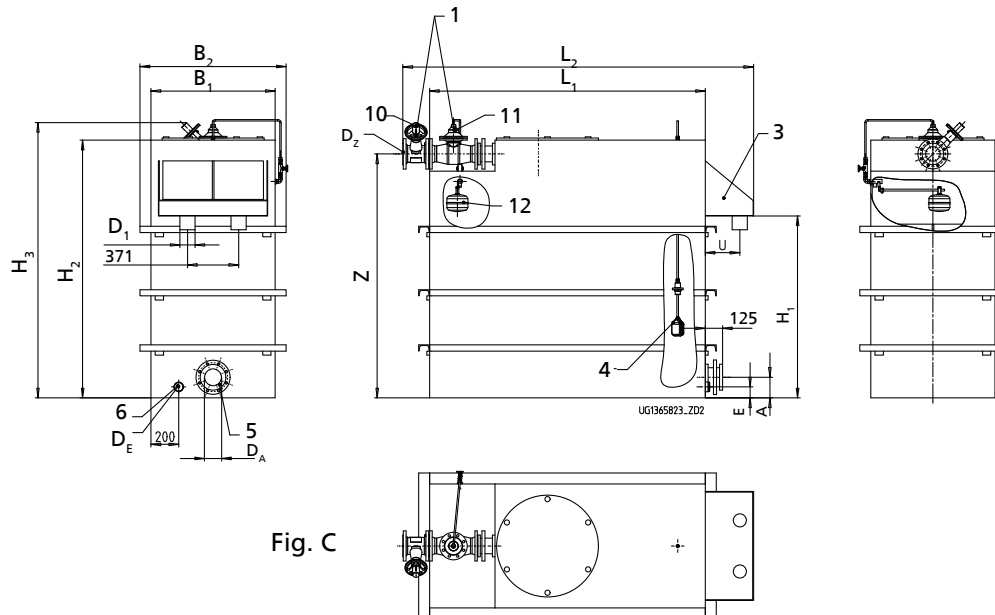


Fig. C

Effective volume of 1500 and 2000 litres

Dimensions [mm]

| Effective volume | Total volume | L <sub>1</sub> | L <sub>2</sub> <sup>5)</sup> | B <sub>1</sub> | B <sub>2</sub> | B <sub>3</sub> | H <sub>1</sub> | H <sub>2</sub> | H <sub>3</sub> | D <sub>1</sub> | D <sub>2</sub> | D <sub>A</sub> | D <sub>E</sub> | Z    | A    | E    | U    | Y    | Fig. |
|------------------|--------------|----------------|------------------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|------|------|------|------|------|------|
| [l]              | [l]          | [mm]           | [mm]                         | [mm]           | [mm]           | [mm]           | [mm]           | [mm]           | [mm]           | [mm]           |                |                |                | [mm] | [mm] | [mm] | [mm] | [mm] |      |
| 150              | 300          | Ø640           | 880                          | -              | -              | 790            | 718            | 1000           | 1095           | 50             | G 1            | DN 50          | G 1            | 920  | 120  | 60   | 90   | 100  | A    |
| 300              | 600          | Ø800           | 1052                         | -              | -              | 950            | 935            | 1250           | 1345           | 75             | G 1 1/2        | DN 50          | G 1            | 1170 | 120  | 60   | 90   | 100  | A    |
| 540              | 750          | Ø800           | 1052                         | -              | -              | 950            | 1184           | 1500           | 1595           | 75             | G 1 1/2        | DN 50          | G 1            | 1420 | 120  | 60   | 90   | 100  | A    |
| 600              | 850          | Ø800           | 1052                         | -              | -              | 950            | 1384           | 1700           | 1795           | 110            | G 1 1/2        | DN 50          | G 1            | 1620 | 120  | 60   | 90   | 100  | A    |
| 800              | 1500         | 1500           | 2012                         | 660            | 740            | 1050           | 1050           | 1524           | 1603           | 2x DN 110      | 2x G 2         | DN 80          | G 1 1/2        | 1445 | 150  | 80   | 250  | 130  | B    |

5) Shut-off valve inlet set included

| Effective volume | Total volume | L <sub>1</sub> | L <sub>2</sub> <sup>5)</sup> | B <sub>1</sub> | B <sub>2</sub> | B <sub>3</sub> | H <sub>1</sub> | H <sub>2</sub> | H <sub>3</sub> | D <sub>1</sub> | D <sub>2</sub> | D <sub>A</sub> | D <sub>E</sub> | Z    | A    | E    | U    | Y    | Fig. |
|------------------|--------------|----------------|------------------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|------|------|------|------|------|------|
| [l]              | [l]          | [mm]           | [mm]                         | [mm]           | [mm]           | [mm]           | [mm]           | [mm]           | [mm]           | [mm]           |                |                |                | [mm] | [mm] | [mm] | [mm] | [mm] |      |
| 1080             | 1800         | 1850           | 2360                         | 660            | 800            | -              | 1050           | 1524           | 1603           | 2x DN 110      | 2x G 2         | DN 80          | G 1 1/2        | 1445 | 150  | 80   | 250  | 130  | B    |
| 1500             | 2700         | 2000           | 2510                         | 900            | 1060           | -              | 1050           | 1524           | 1603           | 2x DN 110      | 2x G 2         | DN 100         | G 1 1/2        | 1445 | 150  | 80   | 250  | 130  | B    |
| 2000             | 3200         | 2000           | 2510                         | 900            | 1060           | -              | 1416           | 1800           | 1975           | 2x DN 110      | 2x G 2         | DN 125         | G 1 1/2        | 1720 | 150  | 80   | 250  | 130  | B    |
| 1500             | 2700         | 2000           | 2550                         | 900            | 1060           | -              | 1038           | 1380           | 1686           | 2x DN 110      | DN 80          | DN 100         | G 1 1/2        | 1490 | 150  | 80   | 250  | 130  | C    |
| 2000             | 3200         | 2000           | 2560                         | 900            | 1060           | -              | 1308           | 1650           | 1998           | 2x DN 110      | DN 100         | DN 125         | G 1 1/2        | 1770 | 150  | 80   | 250  | 130  | C    |

## Installation equipment for Hyamat systems

Installation accessories

| Part No. | Description                                                                                                                                                                                                                                                                                                                                               | Supplementary equipment/option | Accessory mat. No.   | [kg]   | Hyamat K | Hyamat V | Hyamat VP/SVP |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----------------------|--------|----------|----------|---------------|
| 182      |  <p>Foot set for uneven foundations to compensate any differences in height for systems with Movitec 2B / 4B / 6B / 10B / 15B</p> <p><b>Scope of supply:</b><br/>4 feet including fastening elements (2 pumps)<br/>6 feet including fastening elements (3 - 6 pumps)</p> |                                | 18040619<br>18040620 | 2<br>2 | X<br>X   | X<br>X   | X<br>X        |
| 680      |  <p>Acoustic cladding<br/>For systems with Movitec 2B / 4B / 6B / 10B / 15B</p> <p>Protection against contamination<br/>Appealing design<br/>Low-noise operation, approx. 51 - 61 dB(A)</p> <p><b>Designs:</b></p>                                                     |                                |                      |        |          |          |               |
|          | For 2 pumps Movitec 2B / 4B / 6B                                                                                                                                                                                                                                                                                                                          | E113                           | 01117787             | 35     | X        | X        | -             |
|          | For 3 pumps Movitec 2B / 4B / 6B or 2 pumps Movitec 10B / 15B                                                                                                                                                                                                                                                                                             | E114                           | 01117788             | 45     | X        | X        | -             |
|          | For 4 pumps Movitec 2B / 4B / 6B or 3 pumps Movitec 10B / 15B                                                                                                                                                                                                                                                                                             | E115                           | 01117789             | 55     | X        | X        | -             |
|          | For 5 pumps Movitec 2B / 4B / 6B or 4 pumps Movitec 10B / 15B                                                                                                                                                                                                                                                                                             | E116                           | 01117790             | 65     | X        | X        | -             |
|          | For 6 pumps Movitec 2B / 4B / 6B or 5 pumps Movitec 10B / 15B                                                                                                                                                                                                                                                                                             | E117                           | 01117791             | 75     | X        | X        | -             |
|          | For 6 pumps Movitec 10B / 15B (not suitable for Hya-Eco and Hyamat VP)                                                                                                                                                                                                                                                                                    | E118                           | 01117792             | 85     | X        | X        | -             |
| 680      |  <p>Acoustic cladding for systems with Movitec 32 / 45 / 65</p> <p>Protection against contamination<br/>Appealing design<br/>Low-noise operation</p> <p><b>Design up to 7.5 kW</b></p>                                                                                 |                                |                      |        |          |          |               |
|          | with 2 pumps                                                                                                                                                                                                                                                                                                                                              | E123                           | 19071633             | 35     | X        | X        | -             |
|          | with 3 pumps                                                                                                                                                                                                                                                                                                                                              | E124                           | 19071634             | 40     | X        | X        | -             |
|          | with 4 pumps                                                                                                                                                                                                                                                                                                                                              | E125                           | 19071635             | 50     | X        | X        | -             |
|          | with 5 pumps                                                                                                                                                                                                                                                                                                                                              | E126                           | 19071636             | 70     | X        | X        | -             |
|          | with 6 pumps                                                                                                                                                                                                                                                                                                                                              | E127                           | 19071637             | 80     | X        | X        | -             |
|          | <b>Design up to 15 kW</b>                                                                                                                                                                                                                                                                                                                                 |                                |                      |        |          |          |               |
|          | with 2 pumps                                                                                                                                                                                                                                                                                                                                              | E128                           | 19071663             | 35     | X        | X        | -             |
|          | with 3 pumps                                                                                                                                                                                                                                                                                                                                              | E129                           | 19071664             | 40     | X        | X        | -             |
|          | with 4 pumps                                                                                                                                                                                                                                                                                                                                              | E130                           | 19071665             | 50     | X        | X        | -             |
|          | with 5 pumps                                                                                                                                                                                                                                                                                                                                              | E131                           | 19071666             | 70     | X        | X        | -             |
|          | with 6 pumps                                                                                                                                                                                                                                                                                                                                              | E132                           | 19071667             | 80     | X        | X        | -             |

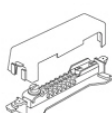
<sup>5)</sup> Shut-off valve inlet set included

Stainless steel baseplates

| Part No. | Description                                                                                                                                                                    | 2-pump Hyamat 2 4 6 | 3-pump Hyamat 2 4 6 | 4-pump Hyamat 2 4 6 | 5-pump Hyamat 2 4 6 | 6-pump Hyamat 2 4 6 | 2-pump Hyamat 10 15 | 3-pump Hyamat 10 15 | 4-pump Hyamat 10 15 | 5-pump Hyamat 10 15 | 6-pump Hyamat 10 15 | 2-pump Hyamat 25 45 65 90 | 3-pump Hyamat 25 45 65 90 | 4-pump Hyamat 25 45 65 90 | 5-pump Hyamat 25 45 65 90 | 6-pump Hyamat 25 45 65 90 | Mat. No. |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|----------|
| 890      | Stainless steel baseplates for pressure booster systems with Movitec 2B, 4B, 6B, 10B, 15B<br> |                     |                     |                     |                     |                     |                     |                     |                     |                     |                     |                           |                           |                           |                           |                           |          |
|          | Baseplate dimensions: [mm]                                                                                                                                                     |                     |                     |                     |                     |                     |                     |                     |                     |                     |                     |                           |                           |                           |                           |                           |          |
|          | 750 x 600 x 60                                                                                                                                                                 | X                   | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                         | -                         | -                         | -                         | -                         | 01348638 |
|          | 980 x 600 x 60                                                                                                                                                                 | -                   | X                   | -                   | -                   | -                   | X                   | -                   | -                   | -                   | -                   | -                         | -                         | -                         | -                         | -                         | 01348639 |
|          | 1210 x 600 x 60                                                                                                                                                                | -                   | -                   | X                   | -                   | -                   | -                   | X                   | -                   | -                   | -                   | -                         | -                         | -                         | -                         | -                         | 01348640 |
|          | 1530 x 600 x 60                                                                                                                                                                | -                   | -                   | -                   | X                   | -                   | -                   | -                   | X                   | -                   | -                   | -                         | -                         | -                         | -                         | -                         | 01348641 |
|          | 1850 x 600 x 60                                                                                                                                                                | -                   | -                   | -                   | -                   | X                   | -                   | -                   | -                   | X                   | -                   | -                         | -                         | -                         | -                         | -                         | 01348642 |
|          | 2170 x 600 x 60                                                                                                                                                                | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                   | X                   | -                         | -                         | -                         | -                         | -                         | 01348643 |
| 890      | Stainless steel baseplates for pressure booster systems with Movitec 25B, 45, 65, 90B<br>     |                     |                     |                     |                     |                     |                     |                     |                     |                     |                     |                           |                           |                           |                           |                           |          |
|          | Baseplate dimensions: [mm]                                                                                                                                                     |                     |                     |                     |                     |                     |                     |                     |                     |                     |                     |                           |                           |                           |                           |                           |          |
|          | 1020 x 600 x 105                                                                                                                                                               | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                   | X                         | -                         | -                         | -                         | -                         | 01359608 |
|          | 1430 x 600 x 105                                                                                                                                                               | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                         | X                         | -                         | -                         | -                         | 01359609 |
|          | 1840 x 600 x 105                                                                                                                                                               | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                         | -                         | X                         | -                         | -                         | 01359610 |
|          | 2250 x 600 x 105                                                                                                                                                               | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                         | -                         | -                         | X                         | -                         | 01359611 |
|          | 2660 x 600 x 105                                                                                                                                                               | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                   | -                         | -                         | -                         | -                         | X                         | 01359612 |

Electrical accessories which can be supplied but not fitted

| Part No.                                                                                   | Description                                                                                                                                                                                                                                                                                                                 | Mat. No. | [kg] | Hyamat K | Hyamat V | Hyamat VP/VP |
|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------|----------|----------|--------------|
|                                                                                            | Alarm switchgear AS 0, AS 2 and AS 4 with circuit breaker, piezoceramic signal transmitter, 85 dB(A) at a distance of 1 m and 4.1 kHz, green equipment-on lamp<br>Plastic housing IP20, 140 x 80 x 57 mm. Use float switch, F1 moisture sensor (item E64), M1 alarm contactor or signal relay of control unit as contactor. |          |      |          |          |              |
| E50<br> | AS 0<br>mains-dependent                                                                                                                                                                                                                                                                                                     | 29128401 | 0.5  | X        | X        | X            |
| E51<br> | AS 2<br>mains-dependent with volt-free signalling contact                                                                                                                                                                                                                                                                   | 29128422 | 0.5  | X        | X        | X            |
| E52<br> | AS 4<br>mains-independent (battery-backed) with volt-free signalling contact and self-charging power supply unit (for 5 hours' operation in the event of power failure)                                                                                                                                                     | 29128442 | 0.5  | X        | X        | X            |

| Part No. | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Mat. No. | [kg]  | Hyamat K | Hyamat V | Hyamat VP/SP |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|----------|----------|--------------|
| E53      |  <p>Alarm switchgear AS 5, mains-independent, with self-charging power supply unit for 10 hours' operation in the event of a mains failure, mains pilot LED, warning indicator light, horn-off push button, volt-free contact for hook-up to a control station, ready for connection with 1.8 m connection cable and plug. ISO housing IP41, 190 x 165 x 75 mm. Use float switch (item E60) as contactor.</p> <p>Horn see accessory E 71</p> <p>230 V ~ / 12 V = 5 VA</p> | 00530561 | 1.7   | X        | X        | X            |
| E64      |  <p>F1 moisture sensor with 3 m cable for alarm switchgear AS0, AS2, AS4</p> <p>Water alarm signal at a water level of only 1 mm (!), by placing the sensor in the outlet section of the inlet tank and/or on the floor of rooms at risk of flooding.</p>                                                                                                                                                                                                                 | 19070212 | 0.3   | X        | X        | X            |
| E70      |  <p>Horn suitable for indoor and outdoor installation, to be mounted in a position where it is protected from direct rain, enclosure IP33</p> <p>12 V DC, 105 dB(A), 1,2 W</p> <p>For use in combination with AS 5</p>                                                                                                                                                                                                                                                    | 01086547 | 0.1   | X        | X        | X            |
| O45      | <p>Plastic housing for easier installation of alarm strobe light (E72), for wall-mounting (WxHxD): 80,5 x 82 x 55 mm</p>                                                                                                                                                                                                                                                                                                                                                                                                                                   | 01061067 | 0.2   | X        | X        | X            |
| E71      |  <p>Alarm combination (alarm strobe light and piezo buzzer), 12 V DC</p> <p>For use in combination with AS 5</p>                                                                                                                                                                                                                                                                                                                                                          | 01139930 | 0.1   | X        | X        | X            |
| E72      |  <p>Alarm strobe light, 12 V DC</p> <p>Enclosure IP 65</p> <p>For use in combination with AS 5</p>                                                                                                                                                                                                                                                                                                                                                                      | 01056355 | 0.3   | X        | X        | X            |
| E73      |  <p>Service software package Automation, customer version</p> <p>CD-ROM with instructions and parameterisation cable (RS232 - MiniUSB) for Windows XP, RS232 interface</p>                                                                                                                                                                                                                                                                                              | 47121211 | 0.4   | X        | X        | X            |
|          | RS232 parameterisation cable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 47117698 | 0.2   | X        | X        | X            |
|          | USB to RS232 interface converter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 01111255 | 0.1   | X        | X        | X            |
|          | Dongle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 47121256 | 0.007 | X        | X        | X            |
| O140     |  <p>Earth rail</p> <p>For additional on-site potential equalisation to VDE0100 Part 410</p> <p>For mounting on the wall or system next to the switchgear</p>                                                                                                                                                                                                                                                                                                            | 01206018 | 2.4   | X        | X        | X            |

Electrical accessories: installation options

| Supplementary equipment/<br>option                                                          | Description                                                                                                                                                                                                      |                                               | Mat. No. | [kg] |          |          |               |
|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|----------|------|----------|----------|---------------|
|                                                                                             |                                                                                                                                                                                                                  |                                               |          |      | Hyamat K | Hyamat V | Hyamat VP/SVP |
| E350<br>   | Manual-0-automatic selector switch per pump<br>For setting the pump to manual mode (non-locking) or idle mode<br>Each pump requires its own manual-0-automatic selector switch.                                  | For systems with 1-6 pumps                    | 19075422 | 0.3  | X        | X        | X             |
| E341<br>   | Voltmeter for each system<br>For displaying the mains voltage of the power supply system<br>In combination with an ammeter                                                                                       |                                               | 19075423 | 0.3  | X        | X        | X             |
| E340<br>   | Ammeter for each pump<br>For displaying the current of each pump<br>In combination with a voltmeter<br>Each pump requires its own ammeter.                                                                       | Measuring range / overload range: 6 A / 12 A  | 19075424 | 0.3  | X        | X        | X             |
|                                                                                             |                                                                                                                                                                                                                  | Measuring range / overload range: 10 A / 20 A | 19075425 | 0.3  | X        | X        | X             |
|                                                                                             |                                                                                                                                                                                                                  | Measuring range / overload range: 15 A / 30 A | 19075426 | 0.3  | X        | X        | X             |
|                                                                                             |                                                                                                                                                                                                                  | Measuring range / overload range: 25 A / 50 A | 19075427 | 0.3  | X        | X        | X             |
|                                                                                             |                                                                                                                                                                                                                  | Measuring range / overload range: 40 A / 80 A | 19075428 | 0.3  | X        | X        | X             |
| E330<br> | Operating hours counter for each pump<br>For displaying the operating hours of each pump<br>Each pump requires its own operating hours counter.                                                                  |                                               | 19075429 | 0.3  | X        | X        | X             |
|          | Volt-free individual fault messages<br>External connection via knife disconnect terminals in the control cabinet                                                                                                 |                                               |          |      |          |          |               |
| E066                                                                                        | For Hyamat K, V, VP                                                                                                                                                                                              | Operation per pump                            | 19075433 | 0.3  | X        | X        | X             |
| E067                                                                                        | For Hyamat K, V                                                                                                                                                                                                  | Fault per pump                                | 19075434 | 0.3  | X        | X        | -             |
| E068                                                                                        | For Hyamat V                                                                                                                                                                                                     | Fault of frequency inverter                   | 19075437 | 0.3  | -        | X        | -             |
| E067                                                                                        | For Hyamat VP                                                                                                                                                                                                    | Fault per pump (frequency inverter)           | 19075439 | 0.3  | -        | -        | X             |
|                                                                                             | "Freely definable" individual signals<br>1 x volt-free contact on isolating terminals for signalling lack of water, excess pressure, insufficient pressure, etc. as required (freely definable via control unit) | Freely definable                              | -        | -    | X        | X        | X             |

| Supplementary equipment/<br>option                                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Mat. No. | [kg]     |          |          |               |   |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|----------|---------------|---|
|                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |          | Hyamat K | Hyamat V | Hyamat VP/SVP |   |
|   | Connection to emergency back-up power supply on second terminal strip<br>For connection to an emergency back-up power supply system which includes mains monitoring and switching between mains network and back-up system<br><br>The control unit does not include mains monitoring and switching between the low voltage distribution system and the generator supply terminal. This can be offered on request.<br>Watch the total power input of the system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 4 kW     | 19075443 | 0.8      | X        | X             | X |
|                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 7.5 kW   | 19075479 | 0.8      | X        | X             | X |
|                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 15 kW    | 19075480 | 1        | X        | X             | X |
|                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 22 kW    | 19075481 | 1        | X        | X             | X |
|                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | >22 kW   | -        | -        | X        | X             | X |
| E059                                                                               | <br>Control cabinet lighting<br>Compact neon light, 11 W, 230 V / 50 Hz - 60 Hz with switch<br>Attaches quickly to DIN rail or to upper control cabinet section by means of magnetic plate.<br>Enclosure IP20<br>Protective glass included in scope of supply.<br><br>Connected before master switch, with separate automatic circuit breaker up to a continuous load of 10 A and earth leakage circuit breaker.<br>For installation in control cabinets of 600 x 400 x 200 mm or larger.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 19075123 | 0.6      | X        | X        | X             |   |
|  | Service socket in the control cabinet<br>Shockproof socket mounted on top hat rail inside the control cabinet Type: 230 V, 10 A (16 A)<br><br>Connected before master switch, with separate automatic circuit breaker and RCD (30 mA).<br>For use as service socket with control cabinet door or outdoor cabinet door open.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 19075122 | 0.4      | X        | X        | X             |   |
| E039                                                                               | <br>Control cabinet heating, 20 W, with hygrostat for specific climatic conditions<br>For using pressure booster system in zones of particularly high humidity or with a risk of condensation in the control cabinet<br><br>The hygrostat controls the control cabinet heating in line with the relative humidity in the housing. When the heater is switched on, the air inside the housing is heated up, the dew point is increased and the relative humidity is reduced, which prevents condensation on assemblies and electronic components.<br><br>Used if the control unit is installed in regions with a high relative humidity, where high temperature fluctuations are to be expected (e.g. pump installations with open reservoirs or in tropical and subtropical climates).<br><br>Technical data: <ul style="list-style-type: none"><li>Control range: 50 - 100 relative humidity</li><li>Switching difference: approx. 4 %</li></ul> Control cabinet size/heater combinations<br><br>1200*1800 = 130 W<br>800*1800 = 90 W<br>800*1200 = 90 W<br>600*1000 = 20 W<br>600*800 = 20 W<br>400*1200 = 20 W<br>600*600 = 20 W<br>400*600 = 20 W | 20 W     | 19075445 | 0.8      | X        | X             | X |
|                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 90 W     | -        |          | X        | X             | X |
|                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 130 W    | -        |          | X        | X             | X |

| Supplementary equipment/<br>option                                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Mat. No. | [kg] |          |          |               |
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|                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |      | Hyamat K | Hyamat V | Hyamat VP/SVP |
|    | <b>E320</b><br>Phase monitoring relay<br>For network monitoring of clockwise rotating field, phase sequence and phase presence (phase failure safety device)<br><br>In the event of a fault in the power supply a message is transmitted to the control unit and a volt-free message is transmitted to the outside via knife disconnect terminals.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 19075448 | 0.8  | X        | X        | X             |
|                                                                                     | <b>E060</b><br> Lightning protection for mains supply, type 1 to EN 61643-11 installed as circuit breaker in the mains supply <ul style="list-style-type: none"> <li>▪ Spark gap based combined lightning current and surge arrester, consisting of base part and plug-in protection modules</li> <li>▪ Can be used up to a supply current of 125 A</li> <li>▪ Maximum system availability by integrated follow current limitation</li> <li>▪ No tripping of 20 A gL/gG fuses up to short-circuit currents of 50 kA</li> <li>▪ Surge arresting capacity up to 100 kA</li> <li>▪ Enables protection of terminal devices</li> <li>▪ Operating state/fault indication by indicator flag in the inspection window</li> <li>▪ Easy replacement of protection modules without tools</li> <li>▪ Vibration and shock tested according to EN 60068-2</li> </ul> | 19075449 | 1.1  | X        | X        | X             |
|  | <b>E061</b><br>Separate control cabinet installation (wall mounting) always without terminal box<br>With 5 m cable for pumps and sensors<br>Maximum length 20 m                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |      |          |          |               |
|                                                                                     | 5 m length per pump 0.37 - 5.5 kW                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 19075450 | 2.2  | X        | X        | X             |
|                                                                                     | 5 m length per pump 7.5 - 15 kW                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 19075451 | 10   | X        | X        | X             |
|                                                                                     | 5 m length per pump 18.5 - 22 kW                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 19075452 | 12   | X        | X        | X             |
|                                                                                     | Terminal box                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 19075453 | 0.3  | X        | X        | X             |

| Supplementary equipment/<br>option                                                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Mat. No. | [kg] |          |          |               |
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|                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |      | Hyamat K | Hyamat V | Hyamat VP/SVP |
| E043<br> | <p>Analog telephone modem for installation in the control cabinet for connection to the general fault message relay or volt-free contacts for individual signals (E31, E32, E51, E66, E67, E68)</p> <p>Analog telephone modem with serial interface and alert transmission, complete with parameterisation cable and TAE connection cable for analog telephone networks</p> <p>Technical data:</p> <ul style="list-style-type: none"> <li>Digital I/O: 8 digital inputs/outputs</li> <li>Analog: 2 inputs (0..20 mA)</li> <li>Analog: 2 outputs (0..10V, at a min. working resistance of 1 kOhm)</li> <li>Interfaces: RS232</li> <li>Transmission of alerts by voice mail, fax, e-mail, text message</li> <li>Remote switching by phone</li> <li>Dimensions (WxHxD): 115 x 120 x 60 mm</li> <li>Top hat rail housing for 35-mm DIN rail</li> <li>Screw terminal connections</li> <li>Operating temperature -20 to + 55 °C, humidity up to 90%, non-condensing</li> <li>Supply voltage 10..30 V DC</li> <li>Current requirement: 150 mA (max.) at 24 V DC</li> <li>TAE cable included</li> <li>Programming cable included</li> </ul> <p>Can be operated via battery-backed IO printed circuit board of LevelControl Basic 2, if rechargeable battery installation option is used.</p> | -        | -    | X        | X        | X             |
| E 041                                                                                     | <p>Profibus field bus module: implementation of Profibus module (supplied fitted) for adjusting the setpoint, alternative setpoint and acknowledging faults via the bus.</p> <p>In addition, all system-relevant parameters can be read.</p> <p>Retrofittable from BoosterControl Advanced (March 2013) Version 1.5 only.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 19074513 | 0    | X        | X        | X             |
| -                                                                                         | <p>Modbus field bus module: implementation of Modbus module (supplied fitted) for adjusting the setpoint, alternative setpoint and acknowledging faults via the bus.</p> <p>In addition, all system-relevant parameters can be read.</p> <p>Retrofittable from BoosterControl Advanced (March 2013) Version 1.5 only.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 19074514 | 0.2  | X        | X        | X             |

| Supplementary equipment/<br>option                                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Mat. No. | [kg] |          |          |               |
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|                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |      | Hyamat K | Hyamat V | Hyamat VP/SVP |
|    | <p>GSM modem, 24 V power supply, without SIM card (SIM card supplied by customer) for connection to the general fault message relay or to the volt-free contacts for individual signals (E31, E32, E51, E66, E67, E68)</p> <p>Without antenna (see below)</p> <p>GSM/GPRS router with serial interface and alert transmission via "virtual leased line"</p> <p>For GSM networks</p> <ul style="list-style-type: none"> <li>▪ Dual-band GSM/GPRS modem</li> <li>▪ EGSM 900 / 1800 MHz</li> <li>▪ Digital I/O: 4 inputs, 2 outputs</li> <li>▪ Analog: 2 inputs</li> <li>▪ Interfaces: RS232</li> <li>▪ Transmission of alerts by fax, e-mail, text message</li> <li>▪ 8 message targets per input in a message chain</li> <li>▪ User-definable shift-change function</li> <li>▪ GPRS Class B</li> <li>▪ Modem: GPRS multislots Class 10 (CS-1 - CS-4)</li> <li>▪ Dimensions (WxHxD): 75 x 120 x 60 mm</li> <li>▪ Top hat rail housing for 35-mm DIN rail</li> <li>▪ Screw terminal connections</li> <li>▪ Operating temperature -20 to + 55 °C, humidity up to 90%, non-condensing</li> <li>▪ Supply voltage 10..30 V DC</li> <li>▪ Current requirement: 150 mA (max.) at 24 V DC</li> <li>▪ Programming cable included</li> </ul> <p>Can be operated via battery-backed IO printed circuit board of LevelControl Basic 2, if rechargeable battery installation option is used.</p> <p>Order processing via Building Service P&amp;A depts. only</p> <p>Additionally required:</p> | -        | -    | x        | x        | x             |
|  | <p>Magnet foot antenna<br/>for D/E network, 0 DB, 2.5-metre cable, FME socket</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | -        | -    | x        | x        | x             |
|  | <p>Omnidirectional antenna<br/>for D/E network, for outdoor mounting, 2.5 DB, 2.5-metre cable, FME socket</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | -        | -    | x        | x        | x             |
|  | <p>Omnidirectional rod antenna<br/>for D/E network, for outdoor mounting, 2 DB, 5-metre cable, FME socket<br/>with stainless steel fixing angle: rod length approx. 30 cm</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | -        | -    | x        | x        | x             |
|  | <p>Also available:<br/>Extension cable for antenna<br/>Cable lengths: 3, 5, 8 or 10 m</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | -        | -    | x        | x        | x             |

| Supplementary equipment/<br>option                                                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Mat. No. | [kg] |          |          |               |
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|                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |      | Hyamat K | Hyamat V | Hyamat VP/SVP |
| E360<br> | Fluid temperature monitoring by contact sensor at the hydraulic part of the pump<br>Adjustable settings<br>Not suitable for 2-stage pumps.<br>Each pump requires its own temperature monitoring device.                                                                                                                                                                                                                                                                                                                 | 19075454 | 0.9  | X        | X        | X             |
| E044<br> | Set of terminals for connecting remote facilities<br>For triggering remote resets, setpoint changeover and operation check runs<br>Cannot be used in combination with the ambient temperature monitoring device.                                                                                                                                                                                                                                                                                                        | 19075456 | 0.3  | X        | X        | X             |
| E041                                                                                      | <b>Profibus field bus module: implementation of Profibus module</b> (supplied fitted)<br>for adjusting the setpoint, alternative setpoint and acknowledging faults via the bus. In addition, all system-relevant parameters can be read.<br>Retrofittable from BoosterControl Advanced (March 2013) Version 1.5 only.                                                                                                                                                                                                   | 19074513 |      | X        | X        | X             |
| -                                                                                         | <b>Modbus field bus module: implementation of Modbus module</b> (supplied fitted)<br>for adjusting the setpoint, alternative setpoint and acknowledging faults via the bus. In addition, all system-relevant parameters can be read.<br>Retrofittable from BoosterControl Advanced (March 2013) Version 1.5 only.                                                                                                                                                                                                       | 19074514 |      | X        | X        | X             |
|        | Insulation monitoring relay for each pump<br>For monitoring the motor insulation<br><br>The monitoring relays emit a warning before the insulation values fall below the critical limits. This serves to detect any damage at an early stage to be able to perform required service work and prevent costly downtimes.<br><br>The motor insulation is monitored in volt-free conditions, with the pump switched off.<br><br>In IT mains insulation monitors must be used in accordance with EN 60204-1 and VDE0100-410. | 19075457 | 0.8  | X        | X        | X             |



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